

Communicating with Models and Data

Hassan Sayed *

October 6, 2026

ROUGH DRAFT

Updated frequently, [click here](#) for latest version.

Abstract

Many real-world decisions rely on external data collection and modeling: consultants provide clients scenarios, fund managers present models of asset returns, and AI chatbots offer recommendations based on internal data. I study how model simplification in these settings distorts not just how data are interpreted but also which data are collected. A sender collects data and estimates a model that helps a receiver make a choice. The sender's model is simple in that it artificially restricts the curvature of the relationship between choices and outcomes: a "naive" model understates curvature and neglects tradeoffs between costs and benefits, while a "cautious" model overstates curvature and exaggerates them. I show that estimating simple models creates strategic incentives to collect data poorly suited to the receiver's decision. Senders who estimate naive models sample data about an excessively large range of choices, while those who estimate cautious models sample locally around a single extreme choice. With multiple variables, senders sample data that maximize collinearity across variables. These distortions make model simplicity strategically valuable: senders who estimate accurate models benefit when receivers misperceive their models as naive or cautious. In equilibrium, accurate senders surround themselves with simple senders who polarize into two groups: those who estimate maximally naive and maximally cautious models. In the limit, these correspond to linear regressions and "worst-case scenario" models, respectively.

*hsayed@cgdev.org, Postdoctoral Scholar, Market Shaping Accelerator. I would like to thank Pietro Ortoleva, Quan Le, Xiaosheng Mu, Kim Sarnoff, Chris Snyder; and audiences at Dartmouth College, the 2026 Stony Brook Game Theory Conference, and the Center for Global Development for productive discussions and feedback.

1 Introduction

Consider a policymaker tasked with designing a carbon tax. She understands the nonlinear trade-off between benefits and costs: low taxes do not curb emissions, while high taxes generate costs. The optimal tax lies between these extremes, but she does not know where. So, she hires a consultant, who collects data and estimates a *model* mapping taxes to welfare. But while his model is valuable, it presents three frictions. First, it may be too simple. Suppose the consultant estimates a linear regression of welfare on taxes. This implies the optimal tax is either very low or high, *neglecting* the tradeoff between benefits and costs. Or, he might estimate a “worst-case scenario” that *exaggerates* the tradeoff by always assuming high costs. In either case, the policymaker believes the model is simplistic, and thus tries to infer the underlying data. This presents a second friction: she cannot observe that data, which may be proprietary or reflect expertise, but which also may be more informative about some taxes than others. Third, the consultant may have financial or reputational incentives to make her optimistic about how much she can improve welfare.

This example reflects a broader phenomenon: decisionmakers rely on others to model the world for them. CEOs hire consultants to guide firm restructuring, investors hire fund managers to model returns, voters use political narratives to understand laws, and, increasingly, any of these decisionmakers may utilize an AI chatbot. A growing literature documents that these models are often *simple* by design. Financial professionals’ “rules of thumb” (Shiller, 2017; Bastianello et al., 2024) and managers’ simplified models of price-setting leave profits on the table (DellaVigna and Gentzkow, 2019; Han et al., 2026). Moreover, their data may be opaque and non-generalizable: experts choose which data to collect (Di Tillio et al., 2021), and AI chatbots may excessively lean on unobservable training data (Jin et al., 2024). Yet these simple models and narratives drive real choices: they may generate optimistic hiring (Flynn and Sastry, 2025), political polarization (Levy et al., 2022; Herrera and Trombetta, 2026), or media engagement (Eliaz and Spiegler, 2026). How do *model simplicity and strategic data collection* shape what decisionmakers learn?

This paper studies a theoretical setting where a receiver (she) hires a sender (he) to help her make a choice. The sender gathers data and estimates a *model* mapping choices to outcomes. His models are *simple* in that they restrict the curvature of the relationship between choices and outcomes to a narrow range of values, rather than letting the data fully determine it. Simple models either *neglect* tradeoffs and understate curvature (such as linear regressions) or *exaggerate* them and overstate curvature (such as worst-case scenarios). I show that a sender who only estimates simple models can strategically sample the data to make the receiver optimistic, since the receiver entertains other models that could have produced those data. If he neglects tradeoffs, he samples data about an excessively wide range of choices. If he exaggerates them, he samples locally around an extreme choice. Then, I show that a sender who estimates an accurate model gains when the receiver mistakes it for a simple one. Accurate senders thus benefit from surrounding themselves with simple ones. These simple senders *polarize* into two groups: ones who fix their models’ curvature near the lowest value the receiver finds plausible, and ones who fix it near the highest. In the limit, these correspond to linear regressions and worst-case scenarios.

Model Overview A receiver — such as a policymaker, CEO, investor, or voter — must choose an action x to maximize an outcome y — such as profits or the value of a public good. She believes that the relationship between y and x is given by a set of data-generating-processes (DGPs) that capture a tradeoff between a linear benefit and a convex cost. The magnitude of the convex cost is given by a parameter $\gamma \in [0, \bar{\gamma}]$, which represents the *curvature* of tradeoffs.

The receiver hires a sender to acquire information for her. The sender chooses a distribution of x values to sample from. He then observes the corresponding y values, producing data (y, x) . His sampling choice is unobserved by the receiver. The sender fits a model to the data. He considers similar models to the receiver, but may be misspecified and treat tradeoffs in a simplified way. I consider two types of senders. The first is a sender who estimates a “naive” model. These models *neglect* tradeoffs the receiver considers by only fitting models that assume low values of curvature γ ; when $\gamma = 0$, the model is a linear regression of y on x . The second type of sender estimates a “cautious” model that *exaggerates* tradeoffs by only assuming high values of γ .

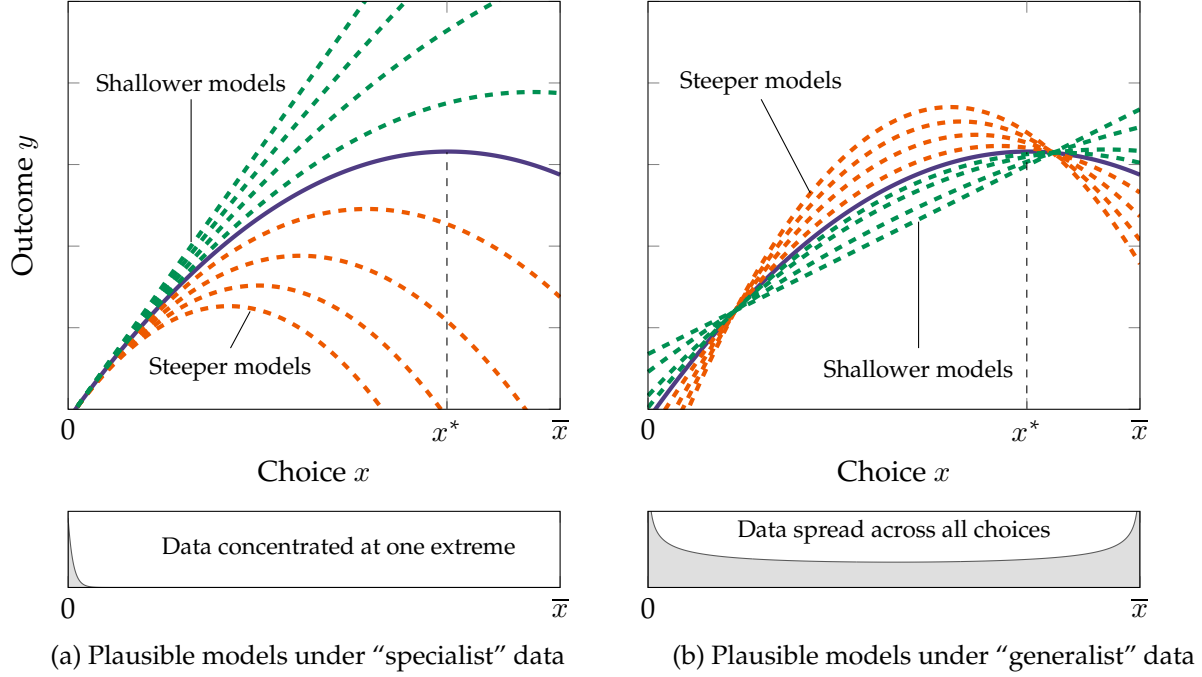
Accordingly, the receiver is Bayesian: she observes the sender’s model, infers how he collected data, and updates over the set of models she considers. The sender’s objective is to make the receiver optimistic, i.e. to make her believe the maximum value of y she can expect to achieve is as *high as possible* — because he would like to maximize her engagement with his services or receives a cut of her expected profits. He knows that the receiver might entertain additional models that will affect her choice, in turn shaping his initial data sampling decision.

Receiver Learning and Sender Sampling The first result is that receiver learning is shaped by the interaction of the sender’s model simplicity with strategic data collection. The sender provides a model, shown in purple in the figure below. However, the receiver may entertain alternative models. These plausible models have *steeper* or *shallower curvature* than the original model.

I highlight two particular forms of data collection. The first is when a sender positions himself as a “specialist” by estimating models “locally” on a small neighborhood of x values. In this case, the receiver’s set of plausible models is tangent to the sender’s model at a single point. This is illustrated in panel (a) of Figure 1. At x^* — where the sender’s model is maximized — steeper models lie *below* the sender’s model and shallower models *above*.

As the set of choices over which the sender samples widens, the receiver’s additional models shift up or down relative to the sender’s model, depending on their curvature. The second form of data collection is when a sender estimates models across the largest possible range of x values. I refer to this as “generalist” sampling. In this case, the models the receiver entertains are *secants* through the sender’s original model. Steeper models lie *above* the original model and shallower models lie *below*. This is shown in panel (b) of Figure 1.

Figure 1: Models receiver entertains when the sender provides simple model



This leads to a second set of results: when the sender presents a *naive* model, he benefits from positioning himself as a generalist. Formally, I show that this sender samples data centered in the middle of the receiver’s choice space with an excessively high variance, which delivers the picture in panel (b) of Figure 1. The intuition is that if the sender presents a naive model, he is neglecting tradeoffs the receiver finds important. This means that the additional models the receiver entertains will be *steeper* than the original model. Steeper models lie above the sender’s original model when the sender positions himself as a generalist. Thus, the sender convinces the receiver she can achieve an even higher expected utility than his original model.

The opposite happens when the sender presents a *cautious model*, where he benefits the most from positioning himself as a specialist. In this case, any additional receiver models are shallower than the original model, which lie above it when he samples as a specialist. Moreover, I show that this sender samples in the smallest possible neighborhood of either the receiver’s maximal or minimal choice. Panel (a) of Figure 1 shows a specialist sender sampling close to $x = 0$. Sampling thus has a “bang-bang” property: the sender either samples data precisely about a fringe receiver choice or across the widest possible range of choices, rather than an intermediate amount.

Third, I extend the baseline model to account for multidimensional choices and show that senders collect data that maximizes collinearity between variables. If a sender neglects a (negative) *interaction* effect between two variables, he samples data so they are as correlated as possible. If he exaggerates interactions, he samples so that those variables are as anticorrelated as possible. The intuition is that if the sender’s model neglects interactions, the sample mean observed in his data is in part driven by a negative interaction of two variables that are highly correlated. If the

receiver were to unwind this correlation, the sample mean would be higher than what she initially observed, meaning her maximal expected utility would also be higher. The opposite logic holds when the sender's model exaggerates interactions. While the paper abstracts from finite samples, this means that senders who estimate simple models collect datasets that are heavily unbalanced, generating imprecise model parameter estimates.

Fourth, I show that a receiver's choices are more accurate under an intermediate degree of sampling, rather than specialist or generalist sampling. Thus, conditional on model simplification, senders do not even sample data in a way that is second-best for the receiver, and include either precise data on choices at the fringe of the receiver's choice space (cautious specialists) or thick-tailed information from all across the choice space (naive generalists). This low quality of data emerges precisely as a consequence of the sender's model simplification.

While most of the paper assumes the receiver's payoff y is quadratic in x , I also show that the results are robust to the receiver's payoffs being given by a linear benefit less a symmetric, convex loss around a point. Although the model without symmetry can prove to be very intractable, I provide a regularity assumption that delivers an analogue of my result: a sender who estimates naive models gains more from being a generalist than a specialist, and one who estimates cautious models gains more from being a specialist than a generalist.

Gains from Simplification and Market for Models Fifth, I show that a sender who provides an accurate model gains from being mistaken as a sender who supplies simple models. I consider a case where one sender always provides an accurate model of choices, and the other provides a simple model. I show that the accurate sender gains from the existence of the simple one. The reason is that, conditional on seeing a model, the receiver believes it is either an accurate model or a simple model. In the latter case, the receiver considers models that deliver higher expected utility than the original, due to the combination of strategic sampling with tradeoff neglect/exaggeration. If a sender who provides an accurate model can plausibly masquerade as one who provides a simple model, he thus makes the receiver more optimistic than he would initially; she considers both his original (true) model and ones that deliver higher expected utility.

Using these insights, I then ask what the *market* for senders may look like for a receiver. This delivers a sixth and final result: whenever a receiver contracts the services of a sender, she will receive one of:

1. an accurate model mapping choices to outcomes;
2. a highly naive model, calibrated to a fixed value of γ close to 0;
3. or a highly cautious model, calibrated to a fixed value of γ close to $\bar{\gamma}$.

To obtain this result, I consider the problem of an accurate sender who always estimates an accurate model of the receiver's choices without simplifying curvature. This accurate sender can surround himself with other senders who may estimate simpler models. For example, a consultant might choose to hire other consultants at his organization who think about tradeoffs in

simpler ways; a politician may be able to propagate simple narratives in the media; or an AI chatbot may deploy simpler models alongside complex ones. In this sense, I allow the accurate sender to choose both the *degree* of simplicity that other senders possess (what models they estimate) as well as their *distribution* in the population (how many of these senders there are). Thus, when the receiver contracts the services of a sender, she must infer whether the model came from the accurate or simple sender.

I show that, in equilibrium, the accurate sender will always choose to surround himself with simple senders who never estimate accurate models, and instead estimate models *pre-calibrated* to a fixed value of γ . For each value of γ , the accurate sender hires a simple sender who can *only* model tradeoffs with that value of curvature. That simple sender then strategically samples data (as a generalist or specialist) so that additional receiver models always lie above the original model. A consultant, for instance, may calibrate a model using predetermined elasticities, rather than estimating one organically on new data.

Then, I show that the *distribution* of these simple sender types is associated with *polarization* in how they consider tradeoffs. I show that the accurate sender always surrounds himself with few (or no) simple senders who model tradeoffs at intermediate values of γ . Instead, he mostly hires senders who estimate models that are (calibrated to be) very cautious (with fixed γ close to $\bar{\gamma}$) or very naive (fixed γ close to 0). These senders not only provide pre-calibrated models, but they tend to be either excessively naive or cautious. In limiting cases, simple senders either *neglect all tradeoffs* and estimate *linear regression models*, or *exaggerate all tradeoffs* and estimate a worst-case model, i.e. those that assume the curvature of tradeoffs is fixed at its maximal value $\bar{\gamma}$.

Overview of Paper Section 2 lays out the basic model. Section 3 characterizes learning, presents the results on data sampling in a model with two sender types, and then provides insights with general sender types and multivariate choices. Section 4 extends the model to investigate the value of learning, forecast error, and non-quadratic costs. Section 5 studies gains from oversimplifying models and the design of the market for senders. Section 6 concludes and discusses implications. The remainder of this section reviews the relevant literature.

Literature This paper broadly adds to our understanding of how we learn from externally-provided models and data, and how incentives to make us optimistic distort the design of information through strategic data collection and model simplification. Its key contribution is to characterize what *sorts* of data model senders may strategically sample, how incentives to strategically sample data are informed by the simplicity of their models' curvature, and how the interaction of both can generate gains from systematically pursuing highly simplistic data-generating-processes.

Many of the examples in the paper concern *delegation* to experts. There is a long line of literature — particularly in organizations and finance — studying this process, including mistrust in experts (Cheng and Hsiaw, 2022), uncertainty about expertise (Kishishita, 2020), and its interaction with persuasion (Kolotilin and Zapechelnyuk, 2025). My contribution to this literature is to

study how delegation and communication operate when an expert’s recommendation comes in the form of a *model* organizing endogenously collected data. The most closely related paper in this literature is Li and Wang (2018), who study delegation when a portfolio manager provides a model of returns to a client.

While there are many papers that study learning from models, much of this literature takes as given the past history of data that a model is fit on — whether that data is exogenous or endogenously generated by a principal’s or agent’s choices (Eliaz and Spiegler, 2020; Schwartzstein and Sunderam, 2021; Levy et al., 2022). My contribution is to study a sender who first chooses a data collection rule and then provides the best-fitting model to that data. Thus, although data collection is endogenous and unobserved by the receiver, the sender’s hands are still tied by his initial data collection. This assumption contrasts with other papers where data is endogenously generated either through follow-on evidence generation (Dai et al., 2026), selective disclosure (Felgenhauer and Schulte, 2014), nonrandom sampling (Di Tillio et al., 2021), or sequential data acquisition with endogenous stopping (Henry and Ottaviani, 2019). These paper show that the ability of a sender to persuade a receiver often depends on the sorts of signals about data a sender can transmit. In my setting, I show that a sender’s desire to make the receiver optimistic coupled with model simplicity creates a tractable characterization of data sampling: the sender either locally samples data in the smallest possible neighborhood of an extreme receiver choice, or as distantly as possible across a wide range of receiver choices.

The use of strategic *sampling* in the present paper in turn interacts with a sender’s model simplicity to make the receiver believe her expected utility is *higher* than under the model provided by the sender. This channel through which a receiver can be made more optimistic is thus related to model simplicity and data sampling, as opposed to departures from a status quo or the inclusion of confounding variables, such as in Eliaz and Spiegler (2020). This also distinguishes the paper from work where a receiver selects between different models with different sample sizes and more or less complex models (Montiel Olea et al., 2022).

My paper shows that gains from model simplicity or strategic sampling can arise without interpreting the role of noise in driving data-generating processes. This finding contrasts with papers that investigate gains from model-based persuasion that involve trading off model fit and belief manipulability — such as Schwartzstein and Sunderam (2021), Aina (2023), Barron and Fries (2024), and Aina and Schneider (2025). The present paper does, however, show that senders who make receivers more optimistic often create greater *forecast* error for a receiver, suggesting that receiver optimism may come at the cost of uncertainty over models. The particular angle the present paper takes is closely related to omitted variable bias — in that a sender cannot (or refuses to) fully model a tradeoff in the way a receiver would like. This connects the paper to work like Levy et al. (2022), Gleyze and Pernoud (2022), and Eliaz et al. (2025) — who show that model misspecification may interact with competitive pressures or uncertainty about data estimation to generate a range of outcomes from improved communication and engagement to political cycles and the prevalence of false models.

A key finding of this paper is that senders may generally benefit from providing *simpler* models to receivers. The idea that senders can benefit from simple models is also captured by Perez-Richet and Prady (2012), Bloedel and Segal (2018), Lipnowski et al. (2020), Eliaz et al. (2021), and Burkovskaya and Starkov (2026). While many of these papers are driven by behavioral assumptions like bounded rationality or rational inattention, the findings in this paper are related to sampling and simplicity: given a sender’s model, a receiver will entertain additional models. The set of these additional models is determined by the *simplicity* of the sender’s model, and the expected utility they generate for the receiver increases or decreases with the spread of the sender’s data sampling. In particular, the present paper shows that a sender informed of the true model of the world can benefit by making the receiver think his model was estimated by a sender who had an artificially simple model and strategically sampled. The most closely related paper is Gleyze and Pernoud (2022), who show that an informed sender can often gain by delegating model provision to a (potentially misspecified) sender.

The present paper is also related to how gains from simplicity operate when a receiver is uncertain about what sort of model she is receiving, which relates it to work such as Dasaratha et al. (2025). Mathematically, senders in the present paper differ in their ability to *model tradeoffs*, which are related to their ability to estimate model curvature on predetermined *subsets* of a parameter space. This characterization of commitment — as well as the problem of the “sender-type designer” — is thus conceptually related to the characterization in Bizzotto et al. (2026) of the limits of commitment in quantity-price games.

Finally, because of its reliance on models as information structures, the present paper also adds to a longer literature on (Bayesian) persuasion (Kamenica and Gentzkow, 2011), as well as persuasion and competition (Gentzkow and Kamenica, 2017; Wang, 2013). In particular, receivers in the present paper are uncertain as to the source of data or the broader class of models a sender entertained. This connects the paper to models of persuasion and cheap talk with *uncertainty* about information structures (Huang, 2024; Min, 2026) and ambiguity aversion (Colo, 2024).

2 Model Setup

I begin by laying out a game between a single receiver R (she) and a single sender S (he). R must make a choice to maximize an outcome, and enlists the services of a sender S . S collects data and provides R a model describing that data. R sees S ’s model, updates, and then makes the choice that maximizes the expected outcome. To help map the model to the real world, I use the running example from the introduction where R is a policymaker tasked with designing a carbon tax, and S is a consultant who models the effects of different taxes.

2.1 Receiver and Models

R faces a set of choices $X = [0, \bar{x}] \subseteq \mathbb{R}$ for $\bar{x} > 0$. Each of these choices maps to an outcome $y \in \mathbb{R}$. Ex ante, R does not know the relationship between y and x . However, she believes that the value of y

given x can be described by a data-generating process (DGP) $y = f_R(x)$ for a function $f_R : X \rightarrow \mathbb{R}$. We refer to f_R as a **model**. We denote F_R as the set of *models* f_R that R entertains. I make the following assumptions on F_R .

Assumption 1. For each $f_R \in F_R$, we have

$$f_R(x) = \alpha_0 + \alpha_1 x - \gamma x^2,$$

where $\alpha_0 \sim P_0^R$ on $\mathbb{R}$, $\alpha_1 \sim P_1^R$ on $\mathbb{R}$, and $\gamma \sim P_\gamma^R$ on $[0, \bar{\gamma}] := \Gamma$ with continuous priors p_0^R, p_1^R, p_γ^R .

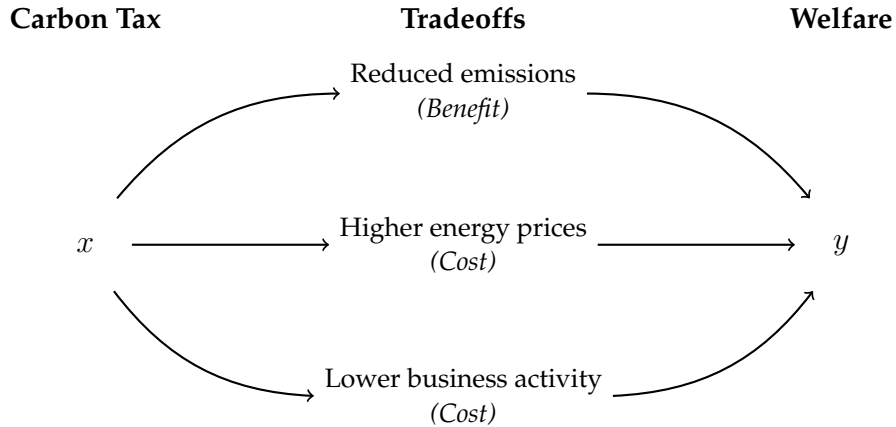
Assumption 1 says that each of R 's functions is concave, and can be represented as an affine function less a convex, quadratic loss.¹ Her prior over each of the three components is independent. Let q denote her belief over F_R , i.e. her (joint) belief over these three parameters. Her expected utility is given by:

$$U_q(x) = \mathbb{E}_q[f_R(x)].$$

Her goal is to choose x to maximize $U_q(x)$.

In our running example, R is a policymaker who designs a carbon tax x . y represents the welfare of a representative household or voter. Higher taxes reduce emissions and improve health outcomes, but may also increase energy prices and reduce private sector activity. The single choice of a tax rate x may have different consequences for welfare, depending on the *aggregate* of these different effects. The policymaker would like to balance these effects and choose the tax rate that maximizes expected welfare. Figure 2 illustrates this idea below.

Figure 2: Effect of x on y through tradeoffs: example of policymaker and consultant



However, there are many other interpretations of the receiver and her preferences. If R is an investor, she cares about how much money she can make by investing in a choice x . If R is a CEO

¹I assume a quadratic form for tractability but show that the central intuitions generalize in Section 4.3.

restructuring her workforce, she cares about the effect of layoffs on profits. If R is a voter, she cares about the impact of a policy x on her wellbeing.²

2.2 Sender and Data

The receiver R contracts a sender S to provide her with information about the relationship between x and y . This sender should be thought of as a *consultant* or *contractor* who provides domain expertise and access to data. Of particular interest is also the case where S is an AI program, such as a large language model (LLM) that R can interface with.

The sender may consider different models than the receiver; which models he considers is determined by a private type. The order of events is as follows.

1. Nature draws S 's type and the receiver's true model f_R .
2. S collects data.
3. S provides a model to R that fits the data he collected. R updates her beliefs about the true data-generating process based on her beliefs about S 's type and the model.
4. R makes a decision x^* . S receives utility. Uncertainty resolves, and R receives her utility.

I now provide more detail about each of these components.

Sender's Type A sender's type affects the simplicity of the models he is capable of providing to R . The key is that S is limited in how he can model the non-affine component of f_R , relative to how R thinks about the problem. The sender's type space is given by $\Theta \subseteq \mathcal{P}(\Gamma)$, and is composed of subsets of $\Gamma = [0, \bar{\gamma}]$. A type θ sender is able to provide R a model in the set F_θ .

$$F_\theta := \{f_\theta \in F_R : f_\theta = \beta_0 + \beta_1 x - \delta x^2, \delta \in \theta, (\beta_0, \beta_1) \in \mathbb{R}^2\}.$$

Intuitively, θ represents the *degree* to which a sender's models artificially *restrict curvature*, that is, can model *tradeoffs*. The sender's type is drawn from a distribution T over Θ known to R .

To start, I assume there are two sender types to highlight the key features of the model. I introduce a richer type space later on.

Assumption 2. $\Theta = \{\theta_1, \theta_2\}$ with $\theta_1 = \{0\}$ and $\theta_2 = \{\bar{\gamma}\}$.

The two sender types in Assumption 2 have the following interpretation. The type θ_1 sender estimates a model $y = \beta_0 + \beta_1 x$, without allowing for any nonlinear effects of x . This sender always understates the curvature of the true DGP (he assumes it is 0). In this sense, he **neglects tradeoffs** and estimates a **naive model**: he runs a linear regression of y on x instead of considering a nonlinear relationship. In our example, the policymaker may believe small carbon taxes generate

²I utilize these simple preferences for higher y in the baseline model, but Section 4.2 extends them to evaluate settings where R also values *forecast accuracy*, and thus knowing the correct model of the world.

meaningful reductions in pollution without adverse effects, but very high taxes lead to higher energy prices or lower business activity. These nonlinear effects would be neglected if a consultant presented a linear regression of welfare on tax rates.

The type θ_2 sender is the opposite. He estimates models of the form $y = \beta_0 + \beta_1 x - \bar{\gamma} x^2$, where $\bar{\gamma}$ is fixed at its highest possible value. This sender always overstates the curvature of the true DGP. We say he **exaggerates tradeoffs** and estimates a **cautious model** of the impact of x on y . This reflects a situation where a consultant estimates a “worst-case scenario.” For example, he may assume that higher taxes always have maximally deleterious effects on the business sector, which the receiver may believe is unlikely.

I assume a type θ sender has a prior over β_0 and β_1 given by measures $\beta_0 \sim P_0^\theta$ and $\beta_1 \sim P_1^\theta$ with support on $\mathbb{R}$. His second-order beliefs about R 's beliefs are as follows: he behaves as if R 's prior over α_0 and α_1 are flat, but correctly believes her prior over γ is given by p_γ^R .³

Sender's Data Prior to choosing a model, S chooses a data sampling strategy μ_θ , where μ_θ is a distribution over X . Intuitively, this means that S is choosing a distribution of choices in X to collect data on. R does not observe S 's choice of μ_θ .

Assumption 3. μ_θ is parametrized by $\mu_\theta(x_\theta, \sigma_\theta) = x_\theta + \sigma_\theta \cdot \epsilon$, where ϵ is mean-zero, symmetric, and $\text{Var}(\epsilon) = 1$. $x_\theta \in X$ and $\sigma_\theta \in [\underline{\sigma}, \bar{\sigma}]$ for $\bar{\sigma} > \underline{\sigma} > 0$ and $\underline{\sigma}$ arbitrarily small.

Assumption 3 provides a convenient parametrization of μ_θ as a distribution centered around a mean x_θ . σ controls the standard deviation of the distribution.⁴ x_θ and σ_θ will denote the equilibrium choices of a type θ sender.

We introduce two definitions here. We say a type θ sender is a **generalist** if $\sigma_\theta = \bar{\sigma}$. Intuitively, a type θ sender's model is estimated *distantly* on as wide a range of x values as possible. We say a type θ sender is a **specialist** if $\sigma_\theta = \underline{\sigma}$. That is, this sender estimates a model *locally* on a small set of x values.

In our example, when the consultant is first contracted by the policymaker, he will choose whether to position himself as a *generalist* or *specialist*. If he is a generalist, then the data he uses to fit his models is drawn from a wide range of carbon tax rates from many contexts. If the consultant is a *specialist*, he will utilize information about a small range of carbon tax rates.

Given this sampling strategy, S directly observes the joint distribution of (y, x) , subject to observational noise. Formally, if the true data-generating process is f_R^* , he observes a joint distribution $D(y, x)$, where the marginal distribution of x is given by $D_x = \mu_\theta$ and the conditional distribution of y given x is given by $D_y(y|x) = f_R^*(x) + \eta$, where η is a mean-zero noise term with unknown variance.⁵

³This is a regularity assumption that shuts down edge-cases of learning in equilibrium.

⁴The constraint that $\sigma > 0$ is a normality assumption made to ensure learning is structured in equilibrium.

⁵I allow S to observe the distribution of data to shut down the role of idiosyncratic data draws in shaping posterior beliefs. This is equivalent to S observing an arbitrarily large (or infinite) number of i.i.d. draws from the true data-generating process.

Sender's Information With the sender's type and sampling strategy in hand, we define the model S provides to R . We say that a model f_S **best-fits the data** with respect to θ and a data-sampling strategy μ_θ if:

$$f_S = \arg \min_{f \in F_\theta} \mathbb{E}_{D(\mu_\theta)}[(y - f(x))^2].$$

For example, when we have $\theta_1 = \{0\}$, f_S represents the *best-fitting linear model* to the data.⁶

I assume S always provides R the best-fitting model to the data. In our running example, the interpretation is that the consultant makes an honest effort to provide R quality information about tax rates, within the constraints of how he approaches the problem. S may use a well-known algorithm to fit his model (such as a linear regression).⁷ This allows me to isolate how model *simplicity* and strategic *data collection* shape learning and sender behavior.

Sender's Preferences Suppose S provides a model $f_S \in F_\theta$. R has a belief over both other plausible models as a function both of f_S as well as the data μ she believes S used. Let x^* be the receiver's choice of $x \in X$ conditional on her posterior beliefs over models q given f_S and her beliefs about μ . A type θ sender's utility is then:

$$U^\theta(x^*) = \mathbb{E}_{q|f_S}^R[f_R(x^*)].$$

Conditional on providing a model f_S , a type θ S would like to maximize R 's posterior expected utility at her chosen action.

The interpretation of these preferences in our example is that the consultant would like to make the policymaker *optimistic*. He may receive a consulting bonus if she is more optimistic about the potential gains she can reap from the carbon tax policy. Relatedly, she may be more likely to utilize the consultant's services in the future. Beyond the current example, however, the preferences lend themselves to several interpretations.

- S is a portfolio manager who receives a fixed share of R 's expected earnings. He cares about the expected profits R can realize (and thus his payment).
- S is a politician who provides information about y to a median voter R . The higher the median voter thinks y is, the more likely she is to turn out in an election. If re-elected, the politician will implement R 's preferred policy x^* .
- S is an AI program that wants to maximize its engagement with its user base. The more optimistic users are, the more likely they will be to purchase a premium subscription.

⁶The existence of a best-fitting model is given by the Hilbert Projection Theorem.

⁷The policymaker could at least partially verify this through a "replication package."

2.3 Additional Comments

In this framework, the receiver’s learning is affected by two key frictions. The first is that the sender’s models are *simple*: S may be unable to think about problems exactly the way R would like. With two types, S either estimates a *naïve* linear model that *totally* neglects tradeoffs, or a *cautious* model that totally exaggerates them and assumes a worst-case scenario. This assumption can be founded through a few different lenses.

1. **Lack of sophistication:** the case where S estimates a naive model is equivalent to one where S estimates a *linear regression* of y on x . The sender may be able to estimate a “line of best fit,” but not understand more complicated model fitting techniques. Thus, he does one of two things. Either he continues to totally neglect nonlinear tradeoffs (because he cannot estimate them), or he overcompensates and assumes a worst-case scenario.
2. **Calibration:** instead of estimating parameters organically on data, S may calibrate his models’ curvature to a fixed value, such as an elasticity from the literature.⁸
3. **Strategic Behavior:** the sender may be able to make the receiver *more optimistic* if she thinks his model was a simplified model. This will be a central result of the paper (Section 5).

The second friction is that the receiver does not know observe the sender’s underlying data and must infer it. For example:

1. The sender may have access to a proprietary dataset, which cannot be directly disclosed to the receiver. Companies like Nielsen, for instance, provide proprietary data on “audience segments” to help advertisers run targeted campaigns (Nielsen, 2024).
2. The sender’s model may also embed *expertise*. The strategy literature suggests that firms often develop internal ways of approaching a problem based on analogous situations they have experienced in the past (Carroll and Sørensen, 2024). Accordingly, a consulting firm’s recommendations may be informed by experience working on cases similar to each other (as a “specialist”) or drawn from many different cases and past contexts (as a “generalist”).
3. The sender is an AI program. It is often difficult for users of these programs (or even their developers) to understand the mapping from internal data to recommendations (Zuccarelli, 2024). In this sense, the mapping from the data to a model is a *black box*.

What matters is that R knows S has access to useful data, but must infer how that data may be strategically sampled when estimating a model.⁹

Additionally, R may also be unable (or unwilling) to expend organizational time and resources in collecting, understanding, and synthesizing complex data. Instead, she delegates the collection

⁸I discuss calibration later when discussing richer type spaces.

⁹ Accordingly, I will be interested in exactly *what sorts of data* the sender will collect in equilibrium, and *how* that data collection interacts with the simplicity of his models.

of data and its synthesis into a model to the sender, expediting her decision-making process in the interest of convenience, but potentially at the cost of information quality.

Finally, I assume that S must provide the best-fitting model to the data, rather than a model that simply “fits the data.” The interpretation is that S makes an honest effort to provide R the best model he can, given the set of data-generating processes he can model. While it could be that S ’s honesty emerges due to professional norms or unmodeled career concerns, it may also be that S synthesizes data using methods or algorithms that are well-known in his field of expertise.¹⁰ Nevertheless, even though S provides a model of best-fit, he can still manipulate the set of models R entertains through the way he initially samples data as well as his type θ .

I focus largely on settings where the nonlinear component of f_R is quadratic. While this is done for tractability, Section 4 discusses the assumptions necessary to generalize the logic.

3 Analysis

This section first investigates the model with the two simple sender types provided above, where one sender provides the receiver a naive “linear regression” model that neglects tradeoffs, and the other a cautious “worst-case” model that exaggerates tradeoffs. I characterize the set of models that the receiver entertains in each case. Using this insight, I show that senders who neglect tradeoffs position themselves as “generalists,” while those who exaggerate them position themselves as “specialists.” I show how this result emerges from the combination of senders’ model simplicity with their ability to strategically sample data.

I then move towards a more complex sender type space to show how the results generalize when senders may only neglect or exaggerate some tradeoffs while correctly estimating others. In either case, I show that the sender’s model is estimated on data that will ex-ante likely be *irrelevant* to the receiver’s eventual choice. When a sender samples as a generalist, he samples in the center of the receiver’s choice space, but includes thick-tailed data about choices at the *edge* of her choice space. If the sender samples as a specialist, he only samples data in a small neighborhood on one of the edges of the receiver’s choice space.

I then further generalize the baseline model with *multidimensional choices*. This delivers an additional prediction: senders may sample data to minimize *collinearity* between different variables. In particular, if a sender *neglects* interaction effects between two variables, he samples data so that they are as *correlated* as possible. If he *exaggerates* interaction effects, he samples data so that they are as *anticorrelated* as possible. Thus, although receivers correctly infer the sort of data that senders collect in equilibrium, that data may be *low quality*, in that it suffers from a high degree of collinearity, and embeds information about choices that will (in expectation) be irrelevant.

3.1 Two Types: Receiver’s Learning and Behavior

I begin by characterizing receiver learning as a function of data-sampling strategies and S ’s type.

¹⁰Relatedly, R could require S ’s methodology to be partially “replicable.”

Receiver's Updating Suppose S provided a model of the form $f_S(x) = \beta_0 + \beta_1 x - \delta x^2$ which best fits the data. In our two-type example, we have either $\delta = 0$ (when $\theta = \theta_1$) or $\delta = \bar{\gamma}$ (when $\theta = \theta_2$). We define $\phi(f_S; \theta, \mu_\theta)$ as the set of functions that could rationalize the data for R , given f_S best-fits with respect to F_θ and μ_θ :

$$\phi(f_S; \theta, \mu_\theta) = \left\{ f_R \in F_R : f_S = \arg \min_{f \in F_\theta} \mathbb{E}_{\mu_\theta} [(f_R(x) - f(x))^2] \right\}.$$

The receiver may entertain additional models beyond f_S . ϕ describes this set. We necessarily have that for each $f_R \in \phi(f_S; \theta, \mu_\theta)$:

$$\frac{\partial}{\partial \beta_i} \mathbb{E}_{\mu_\theta} [(f_R(x) - f(x))^2] = 0,$$

which yields the following moment conditions familiar from ordinary-least-squares (OLS):

$$\mathbb{E}_{\mu_\theta} [f_R(x) - \beta_0 - \beta_1 x + \delta x^2] = 0 \tag{1}$$

$$\mathbb{E}_{\mu_\theta} [x(f_R(x) - \beta_0 - \beta_1 x + \delta x^2)] = 0 \tag{2}$$

Equation (1) is the moment condition with respect to the intercept, and states that the difference between f_R and f_S must average to zero. We note that this is the same as the constraint that a model must “fit the data.” Equation (2) can be combined with equation (1) to read that the covariance between x and $f_R(x) - \beta_0 - \beta_1 x - \delta x^2$ is zero. Both expectations are taken with respect to μ_θ , which is S 's sampling strategy. The following proposition characterizes the set of functions that R entertains given f_S , θ , and μ_θ .

Proposition 1. *Suppose $f_S = \beta_0 + \beta_1 x - \delta x^2$ best-fits the data and $\mu_\theta = \mu_\theta(x_\theta, \sigma_\theta)$. Then, for all $f_R \in \phi(f_S; \theta, \mu_\theta)$, we have:*

$$f_R(x) = f_S(x) + (\gamma - \delta)(\sigma^2 - (x - x_\theta)^2).$$

The proof of this result is contained in the Appendix, and assumes the more general type space Θ introduced later in the section.

Consider the policymaker and consultant. The policymaker can see that the consultant's model is potentially simpler than she would like. Thus, she considers the consultant's original model but also other models that could rationalize the tax-welfare data the consultant saw. Mathematically, the set of models that R considers, given a best-fitting model f_S , takes a simple form: they are all the sender's original model plus a constant, less a quadratic term centered at x_θ . In the case where $\theta_1 = \{0\}$, we have

$$f_R(x) = \beta_0 + \beta_1 x + \gamma(\sigma^2 - (x - x_\theta)^2)$$

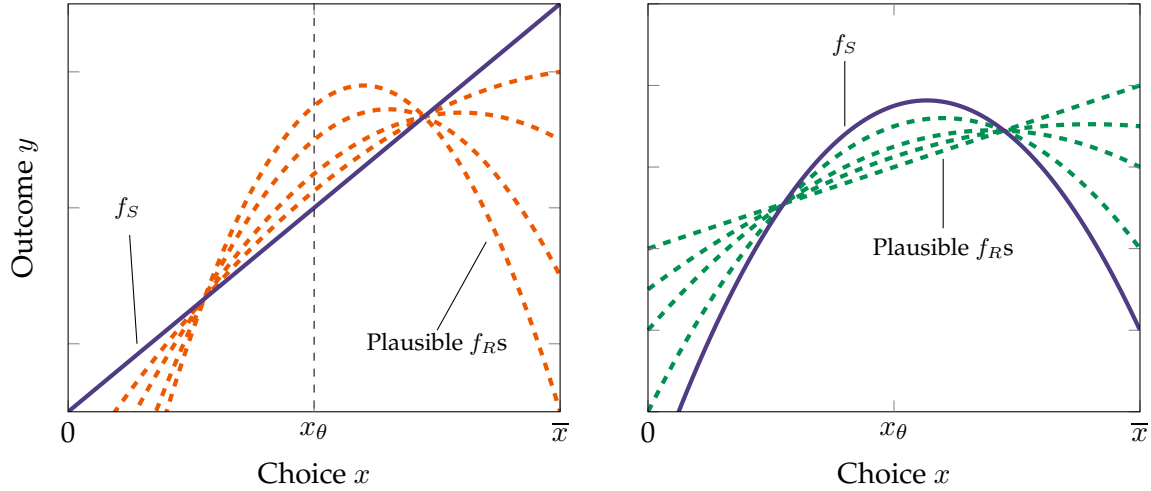
for each $\gamma \in [0, \bar{\gamma}]$. In the case where $\theta_2 = \{\bar{\gamma}\}$, we have

$$f_R(x) = \beta_0 + \beta_1 x - \bar{\gamma} x^2 + (\gamma - \bar{\gamma})(\sigma^2 - (x - x_\theta)^2)$$

for each $\gamma \in [0, \bar{\gamma}]$. In both cases, the set of models R considers involves a fixed “affine component” plus a constant and quadratic term, which may be positive or negative. Moreover, all plausible f_R s have the same slope at x_θ .

This intuition is shown for the type θ_1 in the left panel of Figure 3 below. The purple line shows the sender’s linear regression model. The orange dashed lines indicate all the potential alternative models f_R that the receiver entertains. When S estimates a linear regression model, he neglects any tradeoffs R thinks are important, and thus estimates a relatively naive model. Accordingly, all the additional models that R entertains have a steeper curvature than the regression model, precisely because the linear regression neglects tradeoffs. However, across the sampling range, they have the same sample mean and average slope.

Figure 3: Plausible receiver models f_R as a function of sender model f_S and sampling strategy $\mu_\theta(x_\theta, \sigma_\theta)$



(a) Plausible models when sender provides naive “linear” model (b) Plausible models when sender provides cautious “worst-case” model

The right panel shows the intuition for type $\theta_2 = \{\bar{\gamma}\}$, who estimates a cautious model that exaggerates tradeoffs. Here, all the additional models that R entertains have shallower curvature than f_S . Just like the case with the linear regression, however, they all have the same average slope and sample mean with respect to $\mu_\theta(x_\theta, \sigma_\theta)$.

Receiver’s Decisionmaking The receiver’s learning takes a relatively simple form: regardless of the limitation on S ’s ability, R always learns what the data-generating process looks like “up to the first derivative at x_θ and a constant.” Moreover, as σ changes — and thus S ’s sampling strategy changes — the set of plausible f_R s remains identical, except that they shift up or down

by a constant. This means that R 's optimal *action* depends only on her expectation of γ (given δ and θ). Given f_S , this choice has the following expression.

Corollary 1. *Given $f_S = \beta_0 + \beta_1 x - \delta x^2$, R 's optimal interior action is given by:*

$$x^* = \frac{\beta_1 + 2(\mathbb{E}_\gamma^R[\gamma|f_S] - \delta)x_\theta}{2\delta + 2(\mathbb{E}_\gamma^R[\gamma|f_S] - \delta)}.$$

3.2 Two Types: Sender's Data Sampling and Model Provision

I use these insights to show how S chooses the *spread* of his data as a function of his type. I defer the sender's choice of x_θ to the main statement of the result in Theorem 1.

Proposition 2. *Fix x_θ . The type θ_1 sender who estimates a naive model positions himself as a generalist. The type θ_2 sender who estimates a cautious model positions himself as a specialist.*

The proof of the result is a special case of Theorem 1, which is the version of the result for the more general type space.

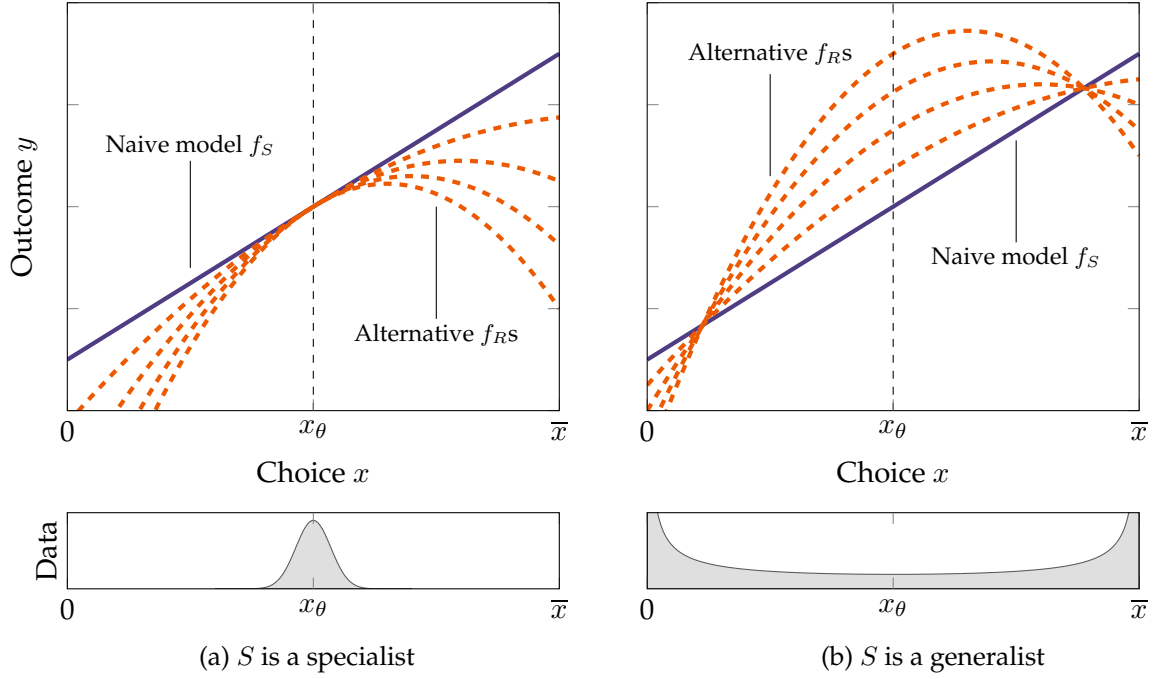
Proposition 2 can be interpreted as follows. If the sender estimates a naive model, the receiver will consider models with $\gamma \geq 0$, i.e. that have *steeper curvature* than the original model. This creates an incentive for him to position himself as a *generalist*, i.e. to estimate his model on as wide a set of choices as possible. However, if the sender estimates a cautious model, the receiver will consider models with $\gamma \leq \bar{\gamma}$, i.e. that have *flatter curvature* than f_S . This gives S an incentive to position himself as a specialist who estimates the DGP locally in a neighborhood of x_θ .

The intuition is driven by the interaction of the curvature of the receiver's additional models with the sender's method of *data sampling*, and is shown visually in Figure 4 below. The top panels (a) and (b) look at the case where f_S is a naive model linear regression of y on x . The bottom panels (c) and (d) look at the case where f_S is the worst-case model fitting the data. The smaller graphs below the graphs of f_S and f_R show what the distribution of *data* underlying different choices is.

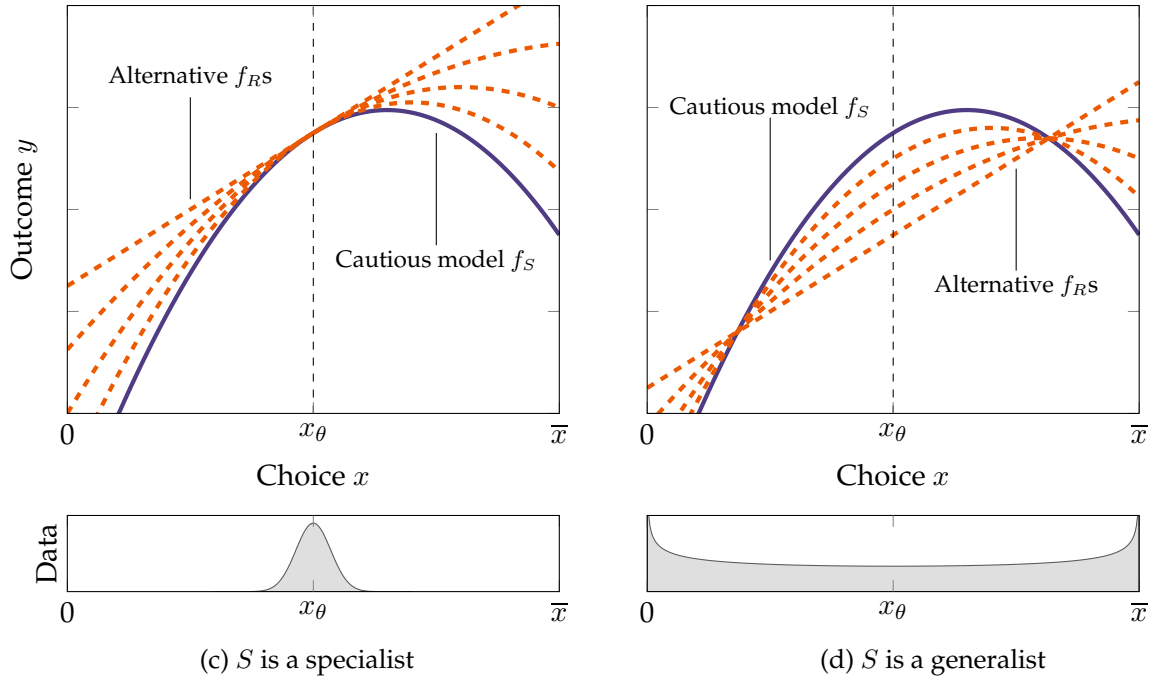
In panel (a), a naive S has positioned himself as a *specialist*, and samples data locally in a neighborhood of x_θ , shown by the shaded gray region. All the additional plausible models f_R are tangent to f_S at x_θ . In panel (a), all these alternative models have steeper curvature (i.e. a higher γ value) than f_S , and lie below f_S at the point of tangency. This logic flips when looking at panel (b), where S has positioned himself as a generalist. Accordingly, the gray region shows that the distribution of data has *spread out* around x_θ . Recall that the distribution of x values when $\sigma = \bar{\sigma}$ is a mean-preserving spread of that when $\sigma = \underline{\sigma}$. To rationalize the data, functions with higher γ values that are "more concave" must shift up relative to shallower ones. In panel (b), this means that the steeper f_R models now lie *above* f_S .

Figure 4: Models that R entertains when S estimates naive vs. cautious models, specialist vs. generalist

(i) S estimates naive “linear” model



(ii) S estimates cautious “worst-case” model



The higher the alternative f_{RS} are above f_S , the more optimistic R is about her prospects.

The logic is flipped in panels (c) and (d). When the sender estimates a “worst-case” model, all the additional models R entertains have shallower curvature (i.e. lower γ) than the sender’s model. When he samples a specialist, all these additional models lie *above* the original model. When he samples as a generalist, these models lie *below* the original model. Thus:

- When S presents naive models, he gains the most by positioning himself as a *generalist*.
- When S presents cautious models, he gains the most by positioning himself as a *specialist*.

We finish this section with a brief commentary on the naive linear regression model. The result on strategic sampling says that this sender always estimates models using data sampled from widely across the choice space. However, regressions are most effective when thought of as *local linear approximations*. Thus, the result on sampling contrasts a sender’s persuasive desire to fit a regression to “distantly sampled data” with the common perception that regressions are most effective when utilized as “local” linear approximations.

3.3 Model with More than Two Types

I now move to generalize the results from the previous section by considering a richer sender type space. In the model with two types, senders either always assumed a naive linear model that completely neglected all tradeoffs, or a cautious worst-case model that exaggerated all of them. In reality, senders and receivers alike may consider multiple tradeoffs in different ways.

- Senders could estimate *some* tradeoffs and assume the worst-case or neglect others. In our example, the consultant may thoroughly estimate how carbon taxes affect energy prices, but assume a worst-case scenario for their effects on political backlash.
- Senders could *calibrate* the value of a tradeoff at a fixed parameter, without estimating it on the real data. For example, the consultant may calibrate the effect of carbon taxes on energy prices using an elasticity from the literature, rather than organically estimating it on data.

Accordingly, I now place the following structure on type spaces.

Assumption 4. For each $\theta \in \Theta$, $\theta \subseteq \Gamma$ is a (potentially degenerate) closed interval. For each $\theta \neq \theta' \in \Theta$, $\theta \cap \theta' = \emptyset$.

Assumption 4 states that each type corresponds to an interval within $[0, \bar{\gamma}]$. I assume type intervals do not intersect for expositional purposes.¹¹ We will denote $\theta = [\underline{\delta}_\theta, \bar{\delta}_\theta]$. Accordingly, type spaces have the following interpretations.

- If $\bar{\delta}_\theta < \bar{\gamma}$, then a type θ S is *neglecting at least some tradeoffs*. Intuitively, these senders do not allow for the case where γ is close to $\bar{\gamma}$.

¹¹The case where they do is of particular interest, and I study it in Section 5.

- If $\underline{\delta}_\theta > 0$, then a type θ S is *exaggerating at least some tradeoffs*. These senders do not allow for the case where γ is close to 0.

Finally, if $\theta = \{\delta\}$ (i.e. θ is a degenerate interval), then a type θ sender **calibrates** his model to a fixed parameter. Intuitively, he assumes γ has a fixed, predetermined value, and does not actually estimate it using his data.

In addition to priors over β_0 and β_1 , a type θ sender now has a prior over δ given by $\delta \sim P_\delta^\theta$ with support on $\theta \subseteq \Gamma$. I make the following assumption on P_δ^θ .

Assumption 5. Suppose $\theta \neq \{0\}$ or $\{\bar{\gamma}\}$. Then, $P_\delta^\theta(\partial\theta \cap (0, \bar{\gamma})) > 0$.

I impose Assumption 5 as a regularity condition; it is consistent with either P_δ being a discrete distribution, or P_δ being a continuous CDF over $[0, \bar{\gamma}]$ whose mass is censored.¹² We denote by $P_{f_S}^\theta$ a type θ sender's prior over models f_S . Second-order beliefs about R 's priors over parameters are the same as before.

Learning with Interval Types The characterization of $\phi(f_S; \theta, \mu_\theta)$ — the set of models R considers in addition to f_S — is the same as above. However, we note that if θ is non-degenerate, we may also have a moment condition corresponding to δ :

$$\mathbb{E}_{\mu_\theta}[x^2(f_R(x) - \beta_0 - \beta_1 x + \delta x^2)] \quad (3)$$

This equation may be zero or nonzero, depending on S 's type. Accordingly, we introduce the following notation:

$$\Delta(f_S) = \mathbb{E}_\gamma^R[\gamma | f_S, \theta] - \delta$$

Fixing f_S , Δ represents the *difference* between R 's expectation of the quadratic parameter γ and the sender's provided parameter δ . In particular, if $\delta \in \theta^\circ$, we necessarily have that (3) is equal to 0, and that f_S is the true model. In this case, $\Delta = 0$. However, if $\delta \in \partial\theta$, then the receiver may entertain additional models.

The statement of Proposition 1 is identical in the case with interval types, and we note that the formal proof utilizes interval types. However, it is useful to state the format of models the receiver considers in the following corollary.

Corollary 2. Consider a sender model $f_S = \beta_0 + \beta_1 x - \delta x^2$ that comes from a type θ sender who sampled data according to a distribution μ_θ . Then, the receiver's expected utility given x is:

$$\mathbb{E}_{f_R | f_S, \theta}[y | x] = f_S(x) + \Delta(f_S)(\sigma_\theta^2 - (x - x_\theta)^2).$$

¹² At a mathematical level, a type θ S 's sampling decision will be determined by how he considers tradeoffs on the “edge” of the interval θ . This “edge” must occur with non-measure-zero probability to have an effect on S 's ex-ante expected utility.

This generalizes the intuitions in the two-type model. If $\delta = \bar{\delta}_\theta < \bar{\gamma}$, then $\Delta(f_S) > 0$: R considers models that have steeper curvature than the one S has provided because he has neglected some tradeoffs. If $\delta = \underline{\delta}_\theta > 0$, then $\Delta(f_S) < 0$: R considers models that have shallower curvature than the one S has provided because he has exaggerated some tradeoffs. In the special case where θ is a singleton, R is completely uncertain as to the value of γ , and $\Delta(f_S)$ may be positive or negative.

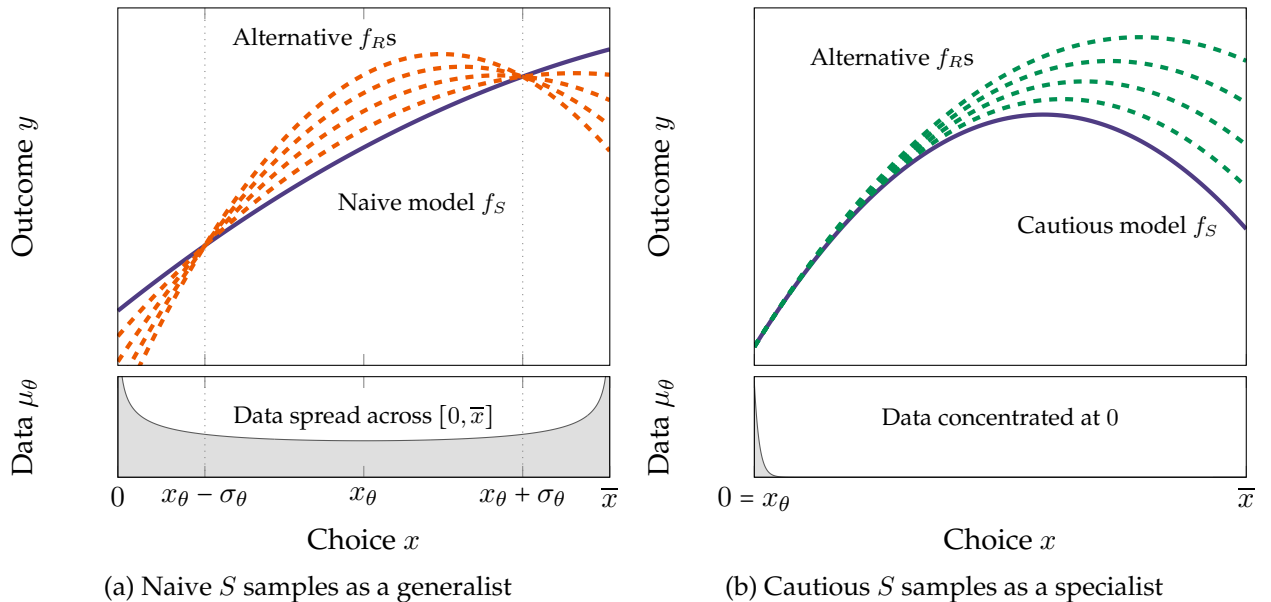
Strategic Sampling with Interval Types We now formalize the following definitions. We say a type θ sender estimates a **naive model** if $\mathbb{E}_{f_S}^\theta(\Delta(f_S)) > 0$. A type θ sender estimates a **cautious model** if $\mathbb{E}_{f_S}^\theta(\Delta(f_S)) < 0$. Concretely, if a sender estimates a naive model, he tends to *neglect* tradeoffs on average, i.e. underestimate γ relative what the receiver considers. If he estimates a cautious model, he tends to *exaggerate* tradeoffs on average, i.e. overestimate γ relative to what the receiver considers.

This allows me to state my main result on strategic sender sampling, which generalizes the logic of Proposition 2.

Theorem 1. *If S estimates naive models, he positions himself as a generalist ($\sigma_\theta = \bar{\sigma}$) and sets $x_\theta = \max\{0, \min\{\mathbb{E}_{f_S}^\theta[x^*(f_S)], \bar{x}\}\}$, where $x^*(f_S)$ is the receiver's choice given f_S . If S estimates cautious models, he positions himself as a specialist ($\sigma_\theta = \underline{\sigma}$) and sets $x_\theta = 0$ or $\bar{x}$.*

The consequence of Theorem 1 is shown in Figure 5 below. If the sender estimates naive models, he positions himself as a generalist, and samples widely around a policy x_θ in the center of the choice space. This is shown on the left. If he estimates cautious models, he positions himself as a specialist, and samples locally near an extreme choice. This is shown on the right panel, where he samples near 0.

Figure 5: Models receiver entertains in equilibrium and data naive vs. cautious sender collects



Theorem 1 can be interpreted as follows. If $\delta \in (\underline{\delta}_\theta, \bar{\delta}_\theta)$, then R believes S 's model is correct. However, if $\delta = \bar{\delta}_\theta$, R will consider models with $\gamma \geq \bar{\delta}_\theta$. Thus, as in the two-type case, this creates an incentive for him to sample from all across the choice space. When $\delta = \underline{\delta}_\theta$, the opposite happens: R considers models with $\gamma \leq \underline{\delta}_\theta$, i.e. “flatter” than those that S provided. Similarly to the two-type case, this gives him an incentive to only estimate models in a small neighborhood of x_θ , i.e. to position himself as a “specialist.”

Theorem 1 additionally shows what value x_θ a sender will choose. The intuition is as follows. When presenting a naive model, S would like to ensure all the additional models that R entertains are as far above f_S as possible. This gap is largest when x_θ is at the maximum of f_S , i.e. where $f'_S(x_\theta) = 0$. Because S does not know ex-ante where this maximum will be, he thus samples where he expects f_S to be the shallowest.

When S estimates a cautious model, the logic is the opposite. If f_S is maximized at a point x , so are all other models tangent to f_S . However, if all these other models are tangent at a point farther from x , the set of tangent models grows higher and higher above f_S . Thus, he samples at either 0 or $\bar{x}$, where he expects f_S to be steepest.

In general, the sender samples data in a way that is *irrelevant* for the receiver's eventual choice. Even though the naive sender's data is centered between 0 and $\bar{x}$, he conditionally makes the receiver more and more optimistic the larger the variance of the data he samples. In practice, this means that he prefers a bimodal distribution with modes close to 0 and $\bar{x}$, rather than a more even single-peaked distribution in the center of the receiver's choice space. This can also be seen visually: the sender's model is *most accurate* at $x_\theta + \sigma$ and $x_\theta - \sigma$ — given by the fact that all the receiver's alternative models are equal to f_S at these points. However, $x_\theta \pm \sigma$ grow arbitrarily large or small with σ . Thus, the naive sender's data suffers from *excessive variance*.

The cautious sender creates the same issue by sampling in the smallest possible neighborhood of 0 or $\bar{x}$. In this sense, he accurately describes the expected value of y given x at just one of these extreme points, but nowhere else. In the example of the policymaker and consultant, this means that a consultant would mostly collect data where a carbon tax is close to 0, or where a carbon tax is excessively high, neither of which is likely to represent the policymaker's optimum.

The coupling of *model simplicity* with *strategic sampling* thus creates incentives to collect *poor data* about the effects of choices on outcomes. Crucially, I note that the receiver, in equilibrium, is not uncertain as to the data the sender collected, and in fact *perfectly* infers what that data are. However, the informational content of those data are ultimately lower than she would like.

3.4 Multidimensional Choices

While the baseline model considered a single choice variable x , I also consider what happens when choices are multidimensional. For example, imagine that our policymaker can shape consumer welfare not just through a carbon tax on businesses but also a complementary clean energy subsidy. I briefly show that many of the findings of the baseline model generalize. However, I also show that when the sender samples data about multiple variables, he always has an incentive to

introduce nearly perfect *collinearity* between them. In particular, if there are two choice variables — x_1 and x_2 — and the sender can choose how *correlated* the two variables are when sampling the data, he will make them either as correlated as possible or as *anticorrelated* as possible.

Multidimensional Models Suppose now that the receiver's choices are given by $x = (x_1, \dots, x_n) \in X^n \subseteq \mathbb{R}^n$. The models f_R the receiver entertains are now as follows.

Assumption 6. For each $f_R \in F_R$, we have

$$f_R(x) = \alpha_0 + \alpha^T x - x^T \gamma x,$$

where $\alpha_0 \sim P_0^R$, $\alpha \sim P_1^R$ for measures on $\mathbb{R}$ and $\mathbb{R}^n$, and $\gamma \sim P_\gamma^R$ on a set Γ . Γ is a compact, convex set of positive-semidefinite matrices such that $0 \in \Gamma$ and such that there exists $\gamma \in \Gamma$ with $\gamma_{ii} > 0$ for all i .

f_R now generalizes the idea of a quadratic data-generating process to a multivariate setting using a set of concave *quadratic forms*. This set of forms is given by a set of positive-semidefinite matrices Γ . Accordingly, I assume the type space Θ of senders is given by a set of compact, convex subsets of Γ .

Assumption 7. A type θ sender provides models of the form:

$$f_S(x) = \beta_0 + \beta^T x - x^T \delta x,$$

where $\delta \in \theta$ for $\theta \subseteq \Gamma$ closed and convex. Moreover, $\theta \cap \theta' = \emptyset$ for all $\theta \neq \theta' \in \Theta$. Additionally, a type θ sender has a prior P_δ^θ over $\delta \in \theta$. Finally, if γ is not an extreme point of Γ , we have $P_\delta^\theta(\partial\theta \cap \Gamma^\circ) > 0$.

Finally, I describe how the sender samples data in multiple variables. As before, I allow the sender to choose a “mean” x_θ and a variance-covariance matrix Σ_θ . Formally, let $\epsilon \in \mathbb{R}^n$ be a *spherically symmetric* random vector that admits a density with $\mathbb{E}[\epsilon] = 0$ and $\text{Cov}(\epsilon) = I_n$. Let Σ_θ be a symmetric positive-definite matrix. Then, if the sender chooses $(x_\theta, \Sigma_\theta)$, his data x are sampled with respect to the distribution $\mu_\theta(x_\theta, \Sigma_\theta)$, such that:

$$\mu_\theta(x_\theta, \Sigma_\theta) = x_\theta + \Sigma_\theta^{\frac{1}{2}} \epsilon.$$

I place the following structure on data sampling.

Assumption 8. $x_\theta \in X^n$. Additionally,

1. $\sigma_i := \sqrt{\Sigma_{\theta, ii}} \in [\underline{\sigma}, \bar{\sigma}]$ and
2. $\rho_{ij} := \frac{\Sigma_{\theta, ij}}{\sigma_i \sigma_j} \in [-\bar{\rho}, \bar{\rho}]$

for $\bar{\rho} \in (0, 1)$ arbitrarily close to 1.

This assumption is a multidimensional generalization of the idea that the “variance” of a variable x_i must be between $\underline{\sigma}^2$ and $\bar{\sigma}^2$. However, the key addition with n choice variables is that the sender can choose the degree of *correlation* between different variables when he samples data. I bound the correlation away from -1 and 1 to ensure the well-definedness of a solution.

Learning with Multidimensional Choices The following result generalizes receiver learning in multiple variables.

Proposition 3. Suppose $f_S = \beta_0 + \beta_1'x - x'\delta x$ best-fits the data and $\mu_\theta = \mu_\theta(x_\theta, \Sigma_\theta)$. Then, for all $f_R \in \phi(f_S; \theta, \mu_\theta)$, we have:

$$f_R(x) = f_S(x) + \text{tr}((\gamma - \delta)\Sigma_\theta) - (x - x_\theta)'(\gamma - \delta)(x - x_\theta).$$

The proposition reads almost identically to Proposition 1, and the intuition is similar as well. However, in place of the original scalar “variance” term, we instead have $\text{tr}((\gamma - \delta)\Sigma_\theta)$. To see the impact that this has on the set of models the receiver will entertain, consider the following example with $x \in \mathbb{R}^2$.

$$\gamma = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}, \quad \delta = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad \Sigma_\theta = \bar{\sigma}^2 \begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix}$$

Here, the receiver’s model allows for a “cross effect” between x_1 and x_2 , while the sender’s model neglects it. In turn, we thus have:

$$\text{tr}((\gamma - \delta)\Sigma_\theta) = \bar{\sigma}^2 \text{tr}\left(\begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix}\right) = \bar{\sigma}^2 \rho.$$

This trace term represents, as before, the *gap* between the sender’s original model and the receiver’s model at x_θ . The sender can *increase* this gap if he makes ρ as large as possible.

More formally, the sender in this case provides models that suffer from *neglecting interactions*. In our running example, the consultant whom the policymaker hires may effectively evaluate the effects of both a subsidy and tax *separately*, but not take into account the *interaction* of their effects.

The receiver is aware of the sender’s misspecification, and she entertains additional models. However, the sender can give the receiver a more optimistic view of these models if he artificially creates a high degree of correlation between x_1 and x_2 when sampling data. In our example, if the consultant cannot figure out how to model these interactive effects, he compensates by skewing his data collection towards settings where both a carbon tax and a clean energy subsidy are active at the same time, rather than towards settings that cleanly isolate the effect of one or the other.

Results with Multidimensional Choices As before, given $f_S = \beta_0 + \beta^T x - x^T \delta x$, we denote $\Delta(f_S)$ as:

$$\Delta(f_S) = \mathbb{E}_\gamma^R[\gamma|f_S] - \delta.$$

The intuitive mapping from the one-dimensional problem to the multidimensional one can be seen by looking at the diagonal entries of $\Delta(f_S)$. If the i^{th} diagonal entry is *positive*, this means that S tends to *neglect tradeoffs* with respect to variable x_i . If it is negative, he tends to *exaggerate* tradeoffs. Thus, we say that a sender estimates a **naive model in x_i** if $\mathbb{E}_{f_S}^\theta[\Delta_{ii}(f_S)] > 0$ and a **cautious model in x_i** if $\mathbb{E}_{f_S}^\theta[\Delta_{ii}(f_S)] < 0$.

However, what happens on the *non-diagonal* entries of Δ is also of interest. Suppose $\Delta_{ij}(\delta) + \Delta_{ji}(\delta) > 0$. This means that the receiver expects the coefficients on the interactions of x_i and x_j to be *larger* than what the sender provides. We say that this sender **neglects interactions between x_i and x_j** . If $\Delta_{ij}(\delta) + \Delta_{ji}(\delta) < 0$, the sender **exaggerates interactions between x_i and x_j** . With these new definitions in hand, we can state the key multidimensional result.

Theorem 2. *Senders who estimate naive models in x_i position themselves as generalists. Those who estimate cautious models in x_i position themselves as specialists. I.e.,*

- if $\mathbb{E}_\delta^\theta[\Delta_{ii}(\delta)] > 0$, then $\sigma_{i,\theta} = \bar{\sigma}$.
- If $\mathbb{E}_\delta^\theta[\Delta_{ii}(\delta)] < 0$, then $\sigma_{i,\theta} = \underline{\sigma}$.

Moreover, senders who neglect interactions between x_i and x_j create collinearity between x_i and x_j . Senders who exaggerate interactions create anticollinearity between x_i and x_j . I.e.,

- if $\mathbb{E}_\delta^\theta[\Delta_{ij}(\delta) + \Delta_{ji}(\delta)] > 0$, then $\Sigma_{ij,\theta} = \bar{\rho}\sigma_{i,\theta}\sigma_{j,\theta}$.
- If $\mathbb{E}_\delta^\theta[\Delta_{ij}(\delta) + \Delta_{ji}(\delta)] < 0$, then $\Sigma_{ij,\theta} = -\bar{\rho}\sigma_{i,\theta}\sigma_{j,\theta}$.

First, Theorem 2 replicates the intuition of the univariate result: if a sender tends to estimate a naive model in variable x_i (i.e. systematically *underestimate* the value of a tradeoff with respect to x_i), he positions himself as a generalist in variable x_i . A cautious sender in variable x_i positions himself as a specialist in that variable.

Theorem 2 also shows what happens when a sender neglects or exaggerates *interactions* between variables. Suppose that our policymaker believes that the interaction of a carbon tax and clean energy subsidy may have particularly deleterious welfare effects at very high values, and that this is totally neglected by the sender. The theorem states that this sender will sample data to favor settings where both a tax and subsidy are active simultaneously, rather than separately.

Intuitively, recall that any additional models the receiver entertains all have the same sample average value of y . Because the tax and subsidy are highly correlated in that data, this sample average is depressed in large part by the (negative) interaction that the sender's model neglects. To rationalize the same sample average, the policymaker's additional models must hence assign

larger separate effects to the tax and subsidy than the sender’s model. But since the tax and subsidy are also choice variables, the receiver need not correlate them in her eventual choice, increasing her maximal expected utility. While this paper does not explicitly deal with finite samples, in practice, the introduction of this collinearity — as well as specialist sampling — can create large standard errors and biased parameter estimates.

In this sense, when choices are multivariate, senders have incentives to sample data in an *unbalanced way*: rather than ensuring that the marginal effects of different variables are estimated independently of each other, there may be an arbitrary degree of collinearity between variables in the datasets that senders produce. Coupled with the previous results where senders collect data about irrelevant choices — either distant data with excessive variance or local data about a fringe receiver choice — high collinearity between variables only worsens the receiver’s ability to learn.

4 Additional Results: Information, Forecasting, General Functional Forms

This section provides additional results from the baseline, univariate model. First, I characterize the *value of information* for the receiver for contracting a sender’s services. I show that although this expression does not depend on whether a sender positions as a generalist or specialist, it potentially *does* depend on *where* the sender sampled data.

Second, I augment the receiver’s baseline preferences with *forecast error*. As a result, it captures settings where the receiver cares not just about maximizing y but also about predicting y accurately. I show that the receiver’s choices are more accurate when she receives information from a specialist than from a generalist. Consequently, the receiver may be better off hiring a cautious sender than a naive sender, although I also show that neither is first best.

Third, I study the model with non-quadratic utility and show that the central intuitions on sender sampling generalize. In particular, under a regularity condition, senders who estimate naive models always prefer to sample data as generalists rather than specialists, and senders who estimate cautious models prefer the opposite. Under an additional symmetry assumption, I am able to obtain the same result as Theorem 1.¹³

4.1 Receiver’s Value of Information

Characterizing the value of information for the receiver from hiring a sender allows us to determine the receiver’s *willingness to pay* to hire the sender, prior to data sampling and model provision. The value of information from hiring a type θ sender is given by:

$$V(\theta) := \mathbb{E}_{f_S|\theta}^R \left[\max_x \mathbb{E}_{f_R|f_S,\theta}^R [f_R(x)|f_S(x)] \right] - \max_x \mathbb{E}_{f_R}^R [f_R(x)] \quad (4)$$

¹³While the proofs of these results are significantly more involved, I also walk through techniques that are utilized in the Appendix to make the problem more tractable.

In the absence of the sender, R would solely rely on her prior to choose x^* . In particular, she would choose $x = \frac{\mathbb{E}_1^R[\alpha_1]}{2\mathbb{E}_\gamma^R[\gamma]}$, which we assume is $\in [0, \bar{x}]$. This allows me to characterize the *value of information* for the receiver as follows.

Proposition 4. Fix θ . The value of information $V(\theta)$ for the receiver from hiring a type θ sender is given by

$$V(\theta) = \mathbb{E}_{f_R}^R \left[\mathbb{E}_\gamma^R [\gamma | f'_R(x_\theta), \delta_\theta(\gamma)] \left(x^* - \frac{\mathbb{E}_1^R[\alpha_1]}{2\mathbb{E}_\gamma^R[\gamma]} \right)^2 \right],$$

where $\delta_\theta(\gamma) := \min\{\max\{\gamma, \underline{\delta}_\theta\}, \bar{\delta}_\theta\}$ and

$$x^* = \max \left\{ \underline{x}, \min \left\{ \bar{x}, x_\theta + \frac{f'_R(x_\theta)}{2\mathbb{E}_\gamma^R[\gamma | f'_R(x_\theta), \delta_\theta(\gamma)]} \right\} \right\}$$

The expression for $V(\theta)$ is familiar from models of information acquisition with quadratic utility. Indeed, if the receiver perfectly learns γ (i.e. $\delta_\theta(\gamma) = \gamma$), $V(\theta)$ collapses to the difference between $\mathbb{E}[\frac{\alpha_1^2}{\gamma}]$ and $\frac{\mathbb{E}[\alpha_1]^2}{\mathbb{E}[\gamma]}$. However, in our setting, learning may be distorted if the sender's model is simple, when the receiver learns f_R up to its first derivative at x_θ and a constant.

Crucially, we note that $V(\theta)$ does not depend on σ , but does depend on x_θ . The intuition for the former is that different sampling strategies cause upwards and downwards shifts in the set of potential models the receiver entertains, which have no bearing on the *choice* the receiver eventually makes.¹⁴

The variation with respect to x_θ is of interest, although finding the optimal value of x_θ proves highly intractable. The following proposition characterizes how $V(\theta)$ varies with x_θ .

Proposition 5. Writing the value of information $V(\theta)$ as a function of the parameter x_θ , $V(\theta, x_\theta)$, we have

$$V_{x_\theta}(\theta, x_\theta) = -\mathbb{E}_{f_R}^R \left[\left(\frac{\partial}{\partial x_\theta} \mathbb{E}_\gamma^R [\gamma | f'_R(x_\theta), \delta_\theta(\gamma)] \right) \left(\frac{f'_R(x_\theta)}{2\mathbb{E}_\gamma^R[\gamma | f'_R(x_\theta), \delta_\theta(\gamma)]} \right)^2 \right].$$

Moreover if p_1^R is absolutely continuous as defined on all of $\mathbb{R}$ and $\lim_{|\alpha_1| \rightarrow \infty} \alpha_1^2 p_1^R(\alpha_1) = 0$, $\arg \max_{x_\theta} V(\theta, x_\theta) > 0$.

Intuitively, x_θ affects the receiver's ability to learn about γ through a *signal extraction problem*. That is, the receiver learns the value of $f'_R(x_\theta) = \alpha_1 - 2\gamma x_\theta$. If $x_\theta = 0$, then the receiver fully learns α_1 and places the same relative weights on different γ values as in her prior. However, if $x_\theta \neq 0$, she learns a weighted combination of α_1 and γ , with the size of the weight determined by x_θ . The nonlinearities introduced by this signal extraction problem make finding an optimal x_θ highly intractable.

¹⁴Generalist and specialist sampling will, however, affect the *variance* of the receiver's outcome, which I address in the next subsection.

4.2 Forecast Accuracy

So far, I have assumed only that the receiver would like to maximize $\mathbb{E}[y|x]$. However, in many settings, the receiver may also directly (or indirectly) value *forecast accuracy*. For example, our policymaker may not just value picking the optimal carbon tax, but also being able to *accurately forecast* the effects of her policy choice. Being able to accurately predict the effects of her policy may allow her to better defend her choice to the press and bureaucracy, as well as to be more confident in her decision.

To illustrate the impact of model simplicity and sampling on forecast-minded receivers, I consider a receiver whose expected utility is given by

$$(1 - \psi)\mathbb{E}^R[y|x] - \psi\text{Var}^R(y|x) \quad (5)$$

for $\psi \in (0, 1)$. Intuitively, these preferences can also be thought of as follows. In a first period, the receiver forms an expectation of y . In a second period, given this expectation, the utility she reaps is based on her ability to forecast. If $\psi = 0$, she does not value the future or forecasting, and only cares about her expectation of y . As $\psi \rightarrow 1$, she places a greater and greater emphasis on forecasting.

Exogenous Forecast Error Before studying what happens when the receiver chooses x to maximize (5), I first characterize how the variance of y behaves with respect to σ_θ when the receiver simply maximizes the expected value of y . The following proposition shows that, fixing x_θ , the variance of outcomes is minimized at an *intermediate* value of σ_θ . That is, neither specialist nor generalist sampling minimizes forecast error for the receiver.¹⁵

Proposition 6. *Fix a sender type θ and let $x^*(f_S)$ maximize $\mathbb{E}_{f_R|f_S, \theta, \mu_\theta}[y|x]$. Then, there exists $\sigma_\theta^* \in (\underline{\sigma}, \bar{\sigma})$ that maximizes*

$$W(\sigma_\theta) := \mathbb{E}_{f_S|\theta}^R[\text{Var}_{f_R|f_S}^R(y|x^*(f_S), x_\theta, \sigma_\theta)].$$

with respect to σ_θ . Moreover, W is strictly decreasing in σ for $\sigma < \sigma_\theta^$ and increasing for $\sigma > \sigma_\theta^*$.*

The intuition for the result is as follows. Writing x^* as the value of x that maximizes the expected value of y , recall our previous formulation that

$$\mathbb{E}[f_R(x^*)|f_S] = f_S(x^*) + \Delta(f_S)(\sigma^2 - (x^*(f_S) - x_\theta)^2)$$

The receiver's forecast error is driven by the latter set of terms multiplied by $\Delta(f_S)$. In particular, the variance of outcomes can be expressed as the difference between the degree of sampling σ^2 and the squared distance between x^* and x_θ (the optimal choice and the point around which data are sampled). When σ^2 is small, the dispersion is driven by $(x^* - x_\theta)^2$, which is always nonzero

¹⁵As before, comparative statics with respect to x_θ are not straightforward, and so we omit them throughout.

in expectation, meaning that the variance can be reduced by increasing σ^2 . Intuitively, because there is uncertainty about what the ex-ante optimal choice x^* will be, an intermediate degree of sampling allows the receiver to “hedge” and gather information in a slightly larger region where she expects the choice to be, rather than precisely at a single point. When σ^2 is very large, however, there is a great deal of vertical dispersion in the potential values of y that may materialize at x^* , and that dispersion can be reduced by decreasing σ^2 .

Thus, the *second-best* degree of sampling σ_θ^* that the receiver would choose to minimize outcome variance is necessarily intermediate: that is, neither a sender who samples as a generalist nor a specialist minimizes the variance of the receiver’s outcomes. However, we note that as $\bar{\sigma}$ grows arbitrarily large, the sender *will* be better off under specialist sampling than generalist sampling.

Endogeneous Forecast Error We now study the setting where R jointly trades off her expected gain in y with forecast error. Conditional on f_S and fixing x_θ , her expected value is

$$V^F(f_S|\theta, \sigma_\theta) = \max_x (1 - \psi) \mathbb{E}_{f_R|f_S, \theta, \sigma_\theta}[y|x] - \psi \text{Var}_{f_R|f_S, \theta, \sigma_\theta}[y|x].$$

We then study the sender’s behavior when the receiver has these preferences.

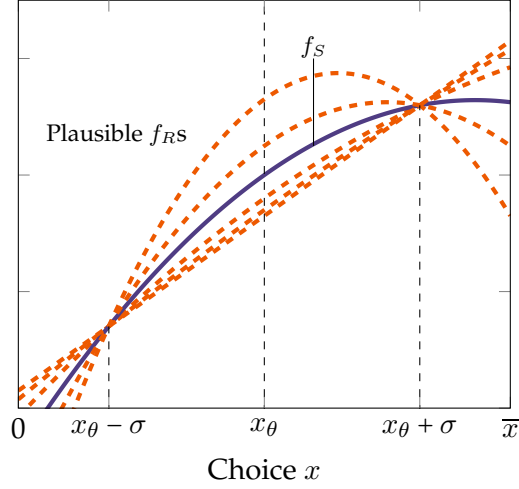
Proposition 7. *Suppose R chooses $x^*(f_S)$ to maximize (5). Then:*

1. *A naive sender chooses $\sigma_\theta > \sigma_\theta^*$ and a cautious sender chooses $\sigma_\theta < \sigma_\theta^*$*
2. *As $\psi \rightarrow 1$, $x^*(f_S) \rightarrow x_\theta \pm \sigma$.*

The first part of the result generalizes the intuition of Theorem 1. Namely, senders who estimate naive models sample data with a *larger spread* than what the receiver would choose, and those who estimate cautious models sample with *less spread*. The intuition is that the sender accounts for the fact that the receiver will shift her choices to take into account forecast error. For very high values of σ^2 , a naive sender trades off between increasing the receiver’s expectation of y through generalist sampling and increasing the variance of outcomes she faces. The opposite logic holds for senders who estimate cautious models.

The second part of the result shows that if the receiver increasingly only values forecast error, her choice approaches $x_\theta \pm \sigma_\theta$. This is illustrated in Figure 6. Thus, R trades off between a choice $x = x^\dagger$ that maximizes her expected utility (at an interior point) and minimizing the variance of outcomes, which happens at $x_\theta \pm \sigma$. When σ is very large, R is forced to choose arbitrarily extreme points $x_\theta \pm \sigma$ to minimize variance. When σ is very small, R chooses points close to x_θ .

Figure 6: Intersection of receiver's models with sender's models



4.3 Non-Quadratic Tradeoffs

While the main model utilizes quadratic utility to express tradeoffs, in this section, I probe the robustness of many of the main results to *non-quadratic* utility. In particular, I show an analogue of Theorem 1: if $\bar{\sigma}$ is sufficiently large and S estimates a naive model, sampling data as a generalist dominates sampling as a specialist. If S estimates a cautious model, being a specialist dominates being a generalist.

General Loss Function The following two assumptions consider two generalizations of functional forms for the receiver.

Assumption 9. *The receiver's models and a type θ sender's models are given by:*

$$f_R(x) = \alpha_0 + \alpha_1 x - \gamma \ell(x - x_\theta) \quad (6)$$

$$f_S(x) = \beta_0 + \beta_1 x - \delta \ell(x - x_\theta) \quad \theta \in \Theta \quad (7)$$

where $\ell : \mathbb{R} \rightarrow \mathbb{R}^+$ is strictly convex and differentiable. Moreover, $\ell(0) = 0$, $\ell'(0) = 0$, and ℓ is symmetric around $x = 0$.

One way to interpret this functional form is to think of a sender's or receiver's model as a *Taylor expansion* around a point x_θ . This Taylor expansion includes a constant component, a linear term, and a concave $\mathcal{O}(2)$ term centered at x_θ . While the convex loss $\ell(x - x_\theta)$ is symmetric around x_θ , the assumption does allow for f_R or f_S themselves to be asymmetric. This assumption subsumes the case where $\ell(x) = x^2$.

The case where this convex loss itself may be locally asymmetric is also of interest. Thus, I also consider the following alternative assumption.

Assumption 10. *The receiver's models and a type θ sender's models are given by:*

$$f_R(x) = \alpha_0 + \alpha_1 x - \gamma m(x) \quad (8)$$

$$f_S(x) = \beta_0 + \beta_1 x - \delta m(x) \quad \theta \in \Theta \quad (9)$$

where $m : \mathbb{R} \rightarrow \mathbb{R}^+$ is strictly convex and differentiable, and $m'(x)$ is onto $\mathbb{R}$. Moreover, there exists x such that $m(x) = m'(x) = 0$.

This formulation of models allows for a convex loss function m to take many different shapes. With these two assumptions in hand, I prove a more general version of Theorem 1.

Theorem 3. *Suppose Assumption 9 holds. If S estimates naive models, he positions himself as a generalist ($\sigma_\theta = \bar{\sigma}$) and sets $x_\theta = \max\{0, \min\{\frac{\mathbb{E}_\theta^\theta[\beta_1]}{2\mathbb{E}_\theta^\theta[\delta]}, \bar{x}\}\}$. If S estimates cautious models, he positions himself as a specialist ($\sigma_\theta = \underline{\sigma}$) and sets $x_\theta = 0$ or $\bar{x}$.*

Alternatively, suppose Assumption 10 holds, suppose $\bar{\sigma}$ is sufficiently large, and fix x_θ . If S estimates naive models, he strictly prefers $\sigma_\theta = \bar{\sigma}$ to $\underline{\sigma}$. If S estimates cautious models, he strictly prefers $\sigma_\theta = \underline{\sigma}$ to $\bar{\sigma}$.

First, we note that the first part of Theorem 3 is identical to Theorem 1: a sender who estimates naive models positions himself as a generalist, and a sender who estimates cautious models positions himself as a specialist. The key reason is that the set of plausible f_R s must obey the following moment conditions.

$$\mathbb{E}_{\mu_\theta}[f_R(x) - \beta_0 - \beta_1 x + \delta \ell(x - x_\theta)] = 0 \quad (10)$$

$$\mathbb{E}_{\mu_\theta}[x(f_R(x) - \beta_0 - \beta_1 x + \delta \ell(x - x_\theta))] = 0 \quad (11)$$

The proof shows that (11) can be boiled down to the condition that the derivative of all plausible f_R s must be the same across the data sample μ_θ . Since $\mu_\theta(x_\theta, \sigma_\theta)$ and $\ell(x - x_\theta)$ are both symmetric around x_θ , we have

$$\mathbb{E}_{\mu_\theta(x_\theta, \sigma_\theta)}[f'_R(x)] = \alpha_1 - \gamma \mathbb{E}_{\mu_\theta(x_\theta, \sigma_\theta)}[\ell'(x - x_\theta)] = \alpha_1 - \gamma \mathbb{E}_{\mu_\theta(x_\theta, \sigma')}[\ell'(x - x_\theta)] \quad \forall \sigma, \sigma' \in [\underline{\sigma}, \bar{\sigma}]$$

Thus, equation (11) is invariant to σ . This means that, as in the quadratic case, as σ changes, the set of models that R considers as a function of σ shifts up or down, allowing us to back out the same logic as in the original case.

However, if models are not characterized by a symmetric loss $\ell(x - x_\theta)$ around x_θ , it is more difficult to obtain a result. Theorem 3 simply states that, fixing x_θ , S prefers being a generalist to being a specialist if he estimates naive models (and the opposite holds if he estimates cautious models). The reason is that, if the convex portion of a receiver's models is *not* symmetric around x_θ , the analogue of (11) no longer holds. I.e., the expression

$$\mathbb{E}_{\mu_\theta(x_\theta, \sigma_\theta)}[f'_R(x)] = \alpha_1 - \gamma \mathbb{E}_{\mu_\theta(x_\theta, \sigma_\theta)}[m(x)]$$

is generally *not invariant* to changes in σ . As a result, the proof allows α_1 to change to “absorb” any changes to $\mathbb{E}_{\mu_\theta(x_\theta, \sigma_\theta)}[m(x)]$ with respect to σ . For each $f_R(x)$ with a given γ value, changing σ thus not only changes α_0 (as in the benchmark and symmetric loss models) but also α_1 . It is here where we utilize the condition in Assumption 10 that assigns the same prior mass to different α_1 values in the receiver’s prior, which ensures that these models receive the same posterior weight.

Consequently, the value of the receiver’s objective at her optimal choice x^* does not have an intuitive expression, and depends on nonintuitive features of the third derivative of $m(\cdot)$. However, the proof shows that if the sender estimates naive models, the value of each $f_R(x)$ the receiver entertains at x_θ is *increasing* in σ . Hence, the value of each f_R at x_θ for sufficiently high $\bar{\sigma}$ must necessarily be higher than the receiver’s optimum x^* at $\underline{\sigma}$. The intuition for the case where the sender estimates cautious models, as before, is flipped.

5 Gains from Simplification

This section returns to the univariate quadratic model to investigate gains from model simplicity. So far, I have taken model simplicity as exogenous: different sender types present models that are artificially naive or cautious. This section characterizes when a sender benefits from the presence of other, simpler senders. To start, I study what happens when senders’ types overlap (i.e., multiple senders can provide the receiver a given model), and characterize when senders can *gain* from the presence of other senders.

Using this result, I then ask what kind of *market* for models a receiver may endogenously face: i.e., how likely she may be to receive an *accurate* or *simple* model, whether simple models tend to be naive or cautious, and how frequent each model type is in the population of senders she faces. In particular, I consider an accurate sender or “designer,” who can always provide an accurate model of the world. However, this sender is able to hire additional senders alongside him who may provide simple models. Conditional on contracting the services of his organization, a receiver may receive information from either the accurate sender or one of these simple senders. Formally: the accurate sender is able to endogenously choose the *type space* of simple senders and their distribution in the population. For example, the accurate sender could be thought of as:

- a consultant or portfolio manager who can employ other consultants/managers whose recommendations may take the form of simple rules of thumb;
- a politician who can recruit co-partisans to propagate simple talking points among voters;
- an AI company that can deploy simpler or more complex LLMs through its chatbot.

The key is that he is able to *shape* the receiver’s beliefs about whether she is facing a simple or accurate model.

I show that the accurate sender would organize the field of senders as follows. First, he would either provide his own services, or those of “simple” senders who provide models that are calibrated to arbitrary parameters. Formally, the type space Θ of simple senders is given by the

collection of singletons $\Theta = \{\{\delta\} : \delta \in [0, \bar{\gamma}]\}$. Second, the *distribution* of these types is characterized by *polarization* in how senders consider tradeoffs. Formally, for each $\delta \in [0, \bar{\gamma}]$, the conditional probability of facing a type $\{\delta\}$ sender is given by a density $t(\delta)$. This density $t(\delta)$ takes the accurate sender's prior over δ and pushes mass away from intermediate values of δ towards the edges of $[0, \bar{\gamma}]$. I.e., simple senders either estimate models with low δ values (close to 0) or high δ values (close to $\bar{\gamma}$), and never estimate models with intermediate δ values. Thus, the set of simple sender types is fully composed of senders who are (arbitrarily) naive or cautious in the models they estimate. As the probability of seeing the accurate sender's model approaches 1, the type space of simple senders converges to a one- or two-point distribution concentrated on either $\delta = 0$, $\delta = \bar{\gamma}$, or both points. These respectively correspond to senders who only provide “linear regression” models or “worst-case scenario” models, as I studied in Section 3.

5.1 Model with Overlapping Types

Previously, I assumed that senders differed in their ability to model the quadratic component of the receiver's models, that different senders could model this component in distinct ways, and thus that the receiver could fully back out the sender's type from the model she received. However, I now relax Assumption 4 and allow the sender's types to have overlapping supports. That is, I consider the case where there exist $\theta, \theta' \in \Theta$ such that $\theta' \cap \theta \neq \emptyset$.

Let $\delta \in \theta \cap \theta'$. Conditional on seeing a model of the form $y = \beta_0 + \beta_1 x - \delta x^2$, the receiver does not know whether this model came from a type θ or type θ' sender. In particular, I can characterize when uncertainty about the source of the signal can actually work to a sender's benefit, and when it hurts the sender.

Proposition 8. *Consider two senders θ, θ' such that $\theta \cap \theta' \neq \emptyset$. Suppose R receives information from either sender with positive probability. Then, a type θ sender's utility either strictly increases or decreases only if $\partial\theta' \cap \theta \neq \emptyset$ or $\partial\theta \cap \theta' \neq \emptyset$.*

There are many permutations to consider, and the proof of the proposition walks exhaustively through all the different cases, but we highlight three here. Denote $\theta = [\underline{\delta}, \bar{\delta}]$ and $\theta' = [\underline{\delta}', \bar{\delta}']$. Given $f_S(x) = \beta_0 + \beta_1 x - \delta x^2$, a type θ sender's utility strictly *increases* from the existence of type θ' in the following cases.

1. $\delta \in \theta^\circ$, $\delta = \bar{\delta}'$, and θ' samples as a generalist.
2. $\delta \in \theta^\circ$, $\delta = \underline{\delta}'$, and θ' samples as a specialist.
3. $\delta = \bar{\delta}$, $\theta' = \{\delta\}$, type θ samples as a specialist, and θ' samples as a generalist.

In the first case, if $\delta \in \theta^\circ$, then f_S is the true model if it came from type θ . However, because $\delta = \bar{\delta}'$, if the model came from θ' , the true model may also have $\gamma \geq \delta$. Accordingly, the type θ' sender samples as a generalist, meaning that R entertains models above f_S . Thus, the receiver

entertains both the possibility that f_S is the true model, but also models with $\gamma \geq \bar{\delta}'$. This benefits the type θ sender, who knows that he has the true model.

The same logic applies to the second case, where $\delta = \underline{\delta}'$ and type θ' samples as a specialist. Type θ provides the true model, but the receiver entertains shallower models that lie above f_S thanks to the type θ' sender's local sampling.

The third case considers a situation where a type θ sender is restricted in how he can sample. In particular, it considers a case where $\delta = \bar{\delta}$ but the type θ sender is forced to sample as a specialist. This may occur because $\underline{\delta}$ is more likely for this sender type, and he thus ex-ante chooses specialist sampling, even though he (ex-post) would be better off sampling as a generalist. However, if $\theta' = \{\delta\}$ and that sender is sampling as a generalist, the type θ sender can be made better off, because the receiver now entertains some models above f_S rather than only ones below.

A corollary of this result is that interactions are zero-sum: if a type θ sender is made better off by a type θ' , the type θ' sender is made worse off.

Corollary 3. *Conditional on providing a model $f_S(x)$, suppose a type θ sender's expected utility increases from a type θ' sender's existence. Then, the type θ' sender's expected utility decreases from the existence of a type θ sender.*

5.2 Constructing the Market for Senders

I now use these intuitions to characterize what sorts of models a receiver may face when she contracts the services of a sender, i.e., what the *market of senders* she experiences looks like. I formalize this problem as an endogenous choice of both a type space Θ and a distribution T over that type space by an accurate sender or “designer.” This sender always provides an accurate model, but can also surround himself with other senders who provide (potentially simpler) models, thus changing the receiver's belief about his accuracy.

Motivation To motivate the role of the accurate sender further, we return to the problem of the consultant and the policymaker. Imagine an “accurate” consultant employed by a consulting organization, who can think about tradeoffs from carbon taxation in *exactly* the way the policymaker would like. However, he can also choose to hire additional colleagues, who might think about tradeoffs in a simpler way than both himself and the policymaker. He can choose how many of these colleagues to hire, as well as how simply they might think about problems.

The consultant thus hires other consultants. The policymaker, in turn, is aware of the composition of the consultant's organization. When she contracts the organization's services, she is randomly assigned either the “accurate” consultant or one of the “simple” consultants. Relatedly, she may contract a team composed of both the accurate consultant as well as other simple consultants, but not know how much oversight the accurate consultant exercised. Based on the model she is ultimately provided, she in turn attempts to *infer* the sender's type. Thus, our problem boils down to: will the accurate consultant hire “simple” consultants to join him, what kind of simplicity will they adopt, and how many of those consultants will he hire?

The key interpretation of the design problem is thus that a accurate sender can endogenously shape the set of senders that the receiver considers, such as by actively hiring additional senders who model tradeoffs differently. Beyond hiring, we also highlight additional interpretations based on our motivating examples.

1. **Portfolio management.** Suppose our sender is a portfolio manager who can estimate a highly accurate model of asset returns. The manager may help train other managers or deploy “robo-advisors” that use simple rules of thumb.
2. **Political narratives.** Suppose our sender is a politician who can accurately provide a “narrative model” of a policy’s effects on voter welfare. However, the same politician can also propagate “simple narratives” of a policy’s effects through online media accounts or by aligning himself with other politicians who provide simple narratives.
3. **AI recommendations.** Suppose our sender is an AI platform that uses a sophisticated LLM to collect data and estimate a robust model that provides users accurate recommendations. The platform can also deploy simpler LLMs that apply a “fixed” way of thinking about a user’s problem, and can determine how often users are shown each model under the hood.

The key mechanism through which this design problem will operate is through manipulating the receiver’s *beliefs* about whether the model she sees came from an accurate or simple sender.

Design Problem Setup Formally, consider the problem of a accurate sender or “designer” D . D has a prior over β_0, β_1, δ as before given by P_0^D, P_1^D, P_δ^D . To ensure the results are not driven by belief asymmetry, I assume that $P_\delta^D = P_\gamma^R$. We write $P_{f_S}^D$ for his prior over models f_S .

I suppose that D faces the following problem. Conditional on a model f_S , with probability q_0 , R believes she has received a model from D . With probability $1 - q_0$, she believes the model came from a type space Θ . D can design the type space Θ . That is, his problem is to choose a set $\Theta \subseteq \mathcal{P}([0, \bar{\gamma}])$ such that for all $\theta \in \Theta$, θ is a closed interval. His objective is:

$$\max_{\Theta, T} q_0 \int_{f_S \in F_R} \max_x \mathbb{E}_{q|f_S, T, \Theta} [f_R(x)] dP_{f_S}^D(f_S).$$

As before, the designer wants to make the receiver optimistic about the expected value of her choice. We note that the impact of Θ is through the receiver’s *posterior*. I take q_0 as fixed; this could represent the designer’s capacity to hire others (higher q_0 means he has less latitude to hire), the exogenous choice of a platform/organization, or a market belief.¹⁶

The order of events is as follows. D first chooses a type space Θ . Then, D chooses a distribution over types T . The receiver contracts the services of a sender, and receives a model from D with probability q_0 or from the type space Θ with probability $1 - q_0$. Conditional on the latter

¹⁶ q_0 could also endogenously be chosen by the designer himself to trade off the probability with which his services will be contracted with receivers’ perceptions of him.

event, a type θ sender is then drawn according to the distribution T . Finally, the deployed sender strategically samples data as before, supplies the model of best fit, and provides it to the receiver.

Optimal Type Space We note that because the designer's models are not misspecified, his data sampling strategy does not affect the alternative models R considers. However, a "simple" sender with type θ will always strategically sample.¹⁷ Thus, the assumption that the receiver cannot directly see the sender's data is key.

In turn, the optimal characterization of the type space is as follows.

Theorem 4. *Suppose P_δ^D is continuous. Then, the optimal type space is the collection of singletons on $[0, \bar{\gamma}] = \Gamma$: $\Theta = \{\{\delta\} : \delta \in \Gamma\}$. Moreover, there exists a function $\pi(\delta)$ such that T is given by a continuous distribution t with*

$$t(\delta) = p_\delta^D(\delta)\pi(\delta). \quad (T^*)$$

In particular, there exist thresholds $0 \leq \underline{\delta}_0 < \mathbb{E}_\gamma^R[\gamma] < \bar{\delta}_0 \leq \bar{\gamma}$ such that $\pi(\delta)$ is strictly decreasing below $\underline{\delta}_0$, strictly increasing above $\bar{\delta}_0$, and equal to 0 on $[\underline{\delta}_0, \bar{\delta}_0]$.

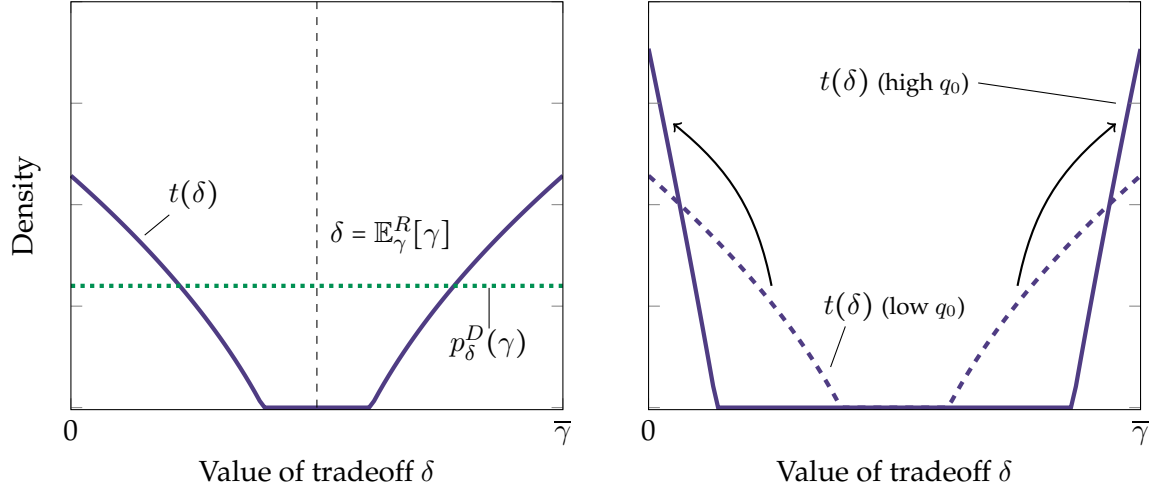
I first interpret the result before describing its mathematical intuition. Theorem 4 says that the optimal type *space* is given by a collection of singletons. For each $\delta \in [0, \bar{\gamma}]$ the accurate sender/designer creates a sender of type $\{\delta\}$. Recalling a prior definition, this sender has the interpretation that he *calibrates* the value of δ to a fixed (but arbitrary) value in $[0, \bar{\gamma}]$. This means that conditional on seeing a model f_S , the receiver believes it is either the true model or was calibrated to an *arbitrary* value.

The second part of the theorem characterizes the *shape* of the distribution of $\{\delta\}$ values the designer utilizes. The solution is that the designer takes his prior over $[0, \bar{\gamma}]$ and *warps* it in a way that pushes more mass to the *edges* of the interval. In this sense, the designer *polarizes* simple senders towards models calibrated to be *arbitrarily* naive (with δ close to 0) or cautious (with δ close to $\bar{\gamma}$). In particular, $t(\mathbb{E}_\gamma^R[\gamma]) = 0$, i.e. t puts 0 mass on types with δ equal to the receiver's prior expectation of γ .

This idea is shown on the left panel of Figure 7 below. The dotted green line shows the *designer's* prior distribution over δ , which is given by a continuous uniform distribution on $[0, \bar{\gamma}]$. The solid purple line shows that the types he creates are concentrated towards the edges at $\delta = 0$ (i.e. linear models) and $\delta = \bar{\gamma}$ (i.e. worst-case models).

¹⁷If the sender is such that the expected value of $\Delta(f_S) = 0$, we assume that sender samples at 0.

Figure 7: Prior distribution $p_\delta^D(\delta)$ of γ vs. optimal distribution of singleton types $t(\delta)$



(a) Optimal distribution of singleton sender types $t(\delta)$ vs. designer's prior over tradeoffs $p_\delta^D(\delta)$ (b) Increased polarization of $t(\delta)$ with increase in likelihood of seeing true model q_0

The first part of the result relies on many of the intuitions of Proposition 8. Consider a model of the form $f_S(x) = \beta_0 + \beta_1 x - \delta x^2$. For each δ , we introduce a sender of type $\{\delta\}$. This sender is simple in the sense that he provides models with an arbitrarily fixed value of δ . Conditional on seeing $f_S(x)$, R would believe that it was either a true model from the accurate sender or the same model provided by the simple type $\{\delta\}$ sender. In the former case, the receiver would believe f_S is the true model. In the latter case, because the simple sender sampled strategically, R 's maximal expected utility from considering alternative models would be higher than from just considering f_S . The designer thus always gains from creating a simple sender for each $\delta \in \Gamma$.

Naturally, D could also introduce a sender of type $\theta = [\underline{\delta}_\theta, \bar{\delta}_\theta]$ with $\theta^\circ \neq \emptyset$. This sender would exaggerate some tradeoffs and neglect others, but also sometimes provide the correct model. In this case, the assumption that P_δ^D has a continuous density is crucial. The only time D could gain from this sender's existence is if that sender provided a model with either $\delta = \bar{\delta}$ or $\underline{\delta}$. However, D would not gain at all if $\delta \in (\underline{\delta}, \bar{\delta})$, since this type θ would always be providing the *true model*. This erases any potential gains from model simplicity and strategic sampling for D . Since $\bar{\delta}$ and $\underline{\delta}$ are single points, each point would have measure 0 in D 's prior, while $(\underline{\delta}, \bar{\delta})$ has measure greater than 0. Thus, D would prefer to have many different senders model tradeoffs arbitrarily, rather than having other senders sometimes provide the true model and sometimes not.

The intuition for the second part of the result is as follows. The accurate sender's problem is to choose a distribution over types which, in this case, boils down to just being a distribution over the set of singletons $\{\delta\}$. In particular, he designates an amount of probability density to allocate to each $\delta \in [0, \bar{\gamma}]$, subject to the constraint that these densities must integrate out to 1.

Suppose $\delta = \mathbb{E}_\gamma^R[\gamma]$, which is the *receiver's* ex-ante expectation of γ . Looking back to Proposition 1, simple senders gain neither from generalist nor specialist sampling in this case. Relatedly, this means that the designer has no incentive to place mass on δ values near $\mathbb{E}_\gamma^R[\gamma]$. On the other hand,

as δ gets farther and farther away from $\mathbb{E}_\gamma^R[\gamma]$, simple senders provide a larger and larger gain from simple models. In fact, the gains to the sender are conditionally maximal if $\delta = 0$ or $\delta = \bar{\gamma}$. This is what drives the “polarization” of the type space.

Variation with q_0 We note that as q_0 increases, the receiver is more likely to think that she saw the accurate sender/designer’s model, and that as q_0 decreases, this is less likely. In the example with the consultant and policymaker, our “designer” may be required to provide his services q_0 of the time, but has latitude to choose the senders who the receiver contracts $1 - q_0$ of the time. As q_0 increases, this latitude decreases, and the receiver is conditionally more and more sure that she received the true model. The following proposition shows that simple sender “polarization” is increasing in q_0 , in the sense that as q_0 increases, $t(\delta)$ places more and more mass on extreme types with very high or low δ values, and less (or none) on intermediate types.

Proposition 9. *The polarization of the type space is increasing in q_0 . That is, $\underline{\delta}_0$ is decreasing in q_0 and $\bar{\delta}_0$ is increasing in q_0 . As $q_0 \rightarrow 1$, $t(\delta)$ converges to a distribution that either places all mass only on $\delta = 0$, $\delta = \bar{\gamma}$, or both. As $q_0 \rightarrow 0$, $t(\delta)$ converges to a distribution that is strictly decreasing on $[0, \mathbb{E}_\gamma^R[\gamma])$, equal to 0 at $\mathbb{E}_\gamma^R[\gamma]$, and strictly increasing on $(\mathbb{E}_\gamma^R[\gamma], \bar{\gamma}]$.*

The proposition is visualized above in the right panel of Figure 7. The dashed purple line graphs $t(\delta)$ for a lower value of q_0 . The solid line graphs $t(\delta)$ for a higher value of q_0 , showcasing the “increased polarization” generated by an increase in q_0 .

The intuition is as follows. The designer’s gain from being mistaken as a simple sender is high at $\delta = 0$, decreases to 0 at $\delta = \mathbb{E}_\gamma^R[\gamma]$, and then rises again until $\delta = \bar{\gamma}$. That is, he conditionally gains the most when δ is very low or very high. When q_0 is large, the ex-ante probability with which he will be seen as a simple sender is low, and he is thus best off concentrating simple senders close to $\delta = 0$ and $\delta = \bar{\gamma}$, where he expects the gain to be largest. However, when q_0 is small, he is willing to distribute simple senders more evenly throughout the population although, in either case, he concentrates few senders near $\delta = \mathbb{E}_\gamma^R[\gamma]$.

I highlight two additional features of this result. First, polarization survives even as $q_0 \rightarrow 0$ (so the receiver is less likely to think she sees the true model). Second, as $q_0 \rightarrow 1$, the optimal distribution either converges to only deploying senders with type $\{0\}$ or type $\{\bar{\gamma}\}$ and nothing in between. That is, the model returns precisely to the two-type case introduced at the beginning of Section 3: in the limit, the only simple senders are those who *only provide* linear regression models or *only provide* “worst-case” models.

6 Conclusion

This paper studies how strategic data collection and model simplicity shape communication with models and data. I study a theoretical model where a receiver R contracts a sender S to help her make a decision. The sender S first collects data and then estimates a model. However, R does

not necessarily know what kind of data S estimated the model on. S 's model may additionally be *simplified* in that it is *naive* and neglects tradeoffs R cares about — i.e. always assumes models have lower curvature — or is *cautious* and exaggerates the extent of tradeoffs — i.e. always assumes models with high curvature. S 's goal is to make R as optimistic as possible. The setting I study has a wide variety of applications to the economics of organizations, portfolio management, political economy, delegation in policymaking, and the use of AI-based decisionmaking.

I first characterize how R learns based on the model that S provides her. I show that R may sometimes believe that S provided an accurate model of her decisions, but that more often than not, R must entertain additional models. If S estimated his model on a small set of choices as a “specialist,” I show that the additional models R entertains are all tangent to S 's original model at a point. If S estimated his model on a large range of choices as a “generalist,” I show that the additional models R entertains are all secants through S 's original model.

I then show a close correspondence between what kinds of data S collects and his model *simplicity*. If S estimates naive models that *neglect* tradeoffs, then he collects data as a generalist across an arbitrarily wide set of a receiver's choices. If S estimates cautious models that *exaggerate* tradeoffs, then he collects data as a specialist locally about a single extreme receiver choice. In both cases, the receiver considers not only the sender's original model, but also models that deliver strictly higher expected utility, creating gains from the coupling of model simplicity and strategic data sampling. Thus, there is a close correspondence between naive models and positioning oneself as a “generalist”; and between cautious models and positioning oneself as a “specialist.” When choices are multivariate, I show that the sender samples data in a way that maximizes collinearity between variables, and makes them as correlated (or anticorrelated) as possible.

Taken together, these points suggest that the quality of *data* may often be *low* when contracting external senders to present model-based recommendations. Moreover, this low quality of data emerges *precisely* as a result of the fact that senders estimate simple models. In particular, the data underlying simple models will often be only loosely related to the choices receivers eventually make. Generalist senders sample data as widely as possible across the choice space and, in particular, place as much mass as possible at its edges. Specialist data are precise, but only describe the value of outcomes at one edge of the choice space. In both cases, the data embed information about choices the receiver is unlikely to end up making. This means that clients should be wary when interpreting the breadth of a consultant's or portfolio manager's data, particularly if they believe that the consultant or portfolio manager did not think about the potential costs of actions as much as they would like. A politician who presents voters a naive “narrative of everything” may only ever be accurate in some cases. On the flipside, even when presented with a “worst-case scenario,” clients should themselves be cautious about whether that data idiosyncratically speaks to an extreme action — such as very high or very low taxes or fringe investment strategies.

The results on multivariate choices show further that the data senders collect may be particularly unbalanced and highly collinear. Although the paper abstracts from finite samples, high collinearity degrades the precision of parameter estimates and the balance of data across covari-

ates, and thus the accuracy of decisions. In our example, a consultant who cannot model how a carbon tax and a clean energy subsidy interact draws evidence from settings where both were adopted together; this leaves the policymaker with little information about what each policy does on its own. More broadly, policymakers may be provided evidence about the effects of multiple reforms operating at once. A firm that hires a strategy consultancy may receive a model based on situations where both advertising campaigns and price promotions were run at the same time. This collinearity makes it difficult to separate the levers that these different choices control.

The paper further shows that the degree of sampling that minimizes forecast error for the sender is an *intermediate* degree of data collection between pure generalism and specialization. Hiring either a specialist or generalist sender is likely to result in imprecise learning, even conditional on a sender's misspecification. This illustrates a tradeoff between optimism and accuracy: the models that make receivers most optimistic do not always lead to the best choices. A CEO who hires the consultant who makes her potential decisions look best may also struggle to pin down the effects of her choices, affecting her ability to justify that decision to a board of directors.

I show that senders who provide the true model can generally gain from the existence of simple senders. That is, if a sender provides a true, accurate model of a receiver's choices, he always gains if the receiver thinks his model was (artificially) naive or cautious. The intuition is that, in the latter cases, the receiver entertains not only a sender's original model but also additional models that lie above it. Thus, the sender who provides an accurate model can make the receiver have a more optimistic belief about her maximal expected utility than the true model. Accurate senders may therefore try to masquerade as simple ones: clients of a large consulting firm may not know whether a senior partner or junior analyst built their model, investors may be unable to tell whether an experienced manager or a robo-advisor suggested a rule, and users of an AI platform may not know whether a recommendation came from a small or a large model.

Finally, I use these results on model simplicity to characterize the market for senders. I consider the problem of an accurate sender who estimates an accurate model of a receiver's tradeoffs. However, he can also surround himself with "simple" senders who estimate naive or cautious models. First, I show that whenever the accurate sender wants to hire simpler senders, he always hires senders who *calibrate* their models to arbitrary parameters, and thus never truly estimate the value of tradeoffs on data. In the real world, these correspond to senders who calibrate their models to pre-defined parameters: economic consultants may calibrate models to elasticities or labor shares from the literature rather than organically estimating them in their setting; robo-advisors may apply fixed investing rules of thumb across clients; and politicians may weave narratives in which a policy's costs are fixed in advance. The paper suggests that calibration of this kind need not reflect a lack of data or expertise, but may instead emerge as a feature of a market for senders that advantages those able to provide the most accurate models.

Second, I show simple senders polarize into two distinct groups: those who estimate (arbitrarily) naive models that are close to "linear regression" models, and those who estimate (arbitrarily) cautious models that are close to "worst-case" models. Thus, the market of senders is composed

of three groups: some senders who always provide accurate models, some who provide highly naive models, and some who provide highly cautious models. In the limit, a receiver sees only linear regressions or worst-case models. These simple models tend to recommend either very extreme or very conservative actions: naive models suggest making choices as extreme as possible, while cautious models may imply sticking to the status quo. While this form of polarization is common among political narratives, the paper suggests that it may occur more generally when an external party is hired to provide a model or *narrative* of the effects of different choices. A market for models may be saturated with models that recommend very extreme or conservative choices.

The combination of simple models, strategic data sampling, and polarization all represent key frictions that may emerge when decisionmakers rely on external expertise and modeling to shape their choices. Future work will investigate these frictions further in the settings of price-based markets and the polarization of political platforms, as well as through empirical work that investigates the use of AI-based decisionmaking in firms or the role of consultants' narratives in shaping organizational decisionmaking.

References

- Aina, C. (2023). Tailored stories. *Unpublished manuscript* 12, 52.
- Aina, C. and F. Schneider (2025). Weighting competing models.
- Barron, K. and T. Fries (2024). Narrative persuasion. Technical report, WZB Discussion Paper.
- Bastianello, F., P. Décaire, and M. Guenzel (2024). Mental models and financial forecasts. *Jacobs Levy Equity Management Center for Quantitative Financial Research Paper*.
- Bizzotto, J., T. Hinnosaar, and A. Vigier (2026). The limits of limited commitment. *American Economic Journal: Microeconomics* 18(1), 254–280.
- Bloedel, A. W. and I. R. Segal (2018). Persuasion with rational inattention. *Available at SSRN* 3164033.
- Burkovskaya, A. and E. Starkov (2026). Causal persuasion.
- Carroll, G. R. and J. B. Sørensen (2024). Strategy theory using analogy: Rationale, tools and examples. *Strategy Science* 9(4), 483–498.
- Cheng, H. and A. Hsiaw (2022). Distrust in experts and the origins of disagreement. *Journal of economic theory* 200, 105401.
- Colo, P. (2024). Communicating about confidence: Cheap talk with an ambiguity-averse receiver. *American Economic Journal: Microeconomics* 16(3), 43–75.
- Dai, Y., D. Fudenberg, and H. Pei (2026). Bayesian persuasion with selective disclosure. *arXiv preprint arXiv:2601.05914*.
- Dasaratha, K., J. Ortner, and C. Zhu (2025). Markets for models. *arXiv preprint arXiv:2503.02946*.
- DellaVigna, S. and M. Gentzkow (2019). Uniform pricing in us retail chains. *The Quarterly Journal of Economics* 134(4), 2011–2084.
- Di Tillio, A., M. Ottaviani, and P. N. Sørensen (2021). Strategic sample selection. *Econometrica* 89(2), 911–953.
- Eliaz, K., S. Galperti, and R. Spiegel (2025). False narratives and political mobilization. *Journal of the European Economic Association* 23(3), 983–1027.
- Eliaz, K. and R. Spiegel (2020). A model of competing narratives. *American Economic Review* 110(12), 3786–3816.
- Eliaz, K. and R. Spiegel (2026). News media as suppliers of narratives (and information). *Theoretical Economics* 21(3), 928–972.

- Eliaz, K., R. Spiegler, and H. C. Thysen (2021). Strategic interpretations. *Journal of Economic Theory* 192, 105192.
- Felgenhauer, M. and E. Schulte (2014). Strategic private experimentation. *American Economic Journal: Microeconomics* 6(4), 74–105.
- Flynn, J. P. and K. Sastry (2025). Contagious business cycles. *Available at SSRN* 4140751.
- Gentzkow, M. and E. Kamenica (2017). Bayesian persuasion with multiple senders and rich signal spaces. *Games and Economic Behavior* 104, 411–429.
- Gleyze, S. and A. Pernoud (2022). The value of model misspecification in communication.
- Han, Y., D. Huffman, and Y. Liu (2026). Minds, models and markets: How managerial cognition affects pricing strategies.
- Henry, E. and M. Ottaviani (2019). Research and the approval process: The organization of persuasion. *American Economic Review* 109(3), 911–955.
- Herrera, H. and F. Trombetta (2026). Alternative worldviews, distrust, and populism. *Distrust, and Populism* (February 23, 2026).
- Huang, W. (2024). Learning about informativeness. *arXiv preprint arXiv:2406.05299*.
- Jin, Z., J. Liu, Z. Lyu, M. Sachan, R. Mihalcea, M. Diab, D. Ha, et al. (2024). Can large language models infer causation from correlation? In *International Conference on Learning Representations*, Volume 2024, pp. 28663–28679.
- Kamenica, E. and M. Gentzkow (2011). Bayesian persuasion. *American Economic Review* 101(6), 2590–2615.
- Kishishita, D. (2020). (not) delegating decisions to experts: The effect of uncertainty. *Journal of Economic Theory* 190, 105117.
- Kolotilin, A. and A. Zapechelnyuk (2025). Persuasion meets delegation. *Econometrica* 93(1), 195–228.
- Landsman, Z. and J. Nešlehová (2008). Stein’s lemma for elliptical random vectors. *Journal of Multivariate Analysis* 99(5), 912–927.
- Levy, G., R. Razin, and A. Young (2022). Misspecified politics and the recurrence of populism. *American Economic Review* 112(3), 928–962.
- Li, Y. and C. Wang (2018). Delegation uncertainty. *Columbia Business School Research Paper* (15-54).
- Lipnowski, E., L. Mathevet, and D. Wei (2020). Attention management. *American Economic Review: Insights* 2(1), 17–32.

- Min, W. (2026). Information acquisition with uncertain signal structure. *Available at SSRN 5118651*.
- Montiel Olea, J. L., P. Ortoleva, M. M. Pai, and A. Prat (2022). Competing models. *The Quarterly Journal of Economics* 137(4), 2419–2457.
- Nielsen (2024). Audience targeting solutions. <https://www.nielsen.com/solutions/media-planning/audience-segments/>. Accessed 2026-09-03; page last modified 2024-09-12.
- Perez-Richet, E. and D. Prady (2012). Complicating to persuade?
- Schwartzstein, J. and A. Sunderam (2021). Using models to persuade. *American Economic Review* 111(1), 276–323.
- Shiller, R. J. (2017, April). Narrative economics. *American Economic Review* 107(4), 967–1004.
- Wang, Y. (2013). Bayesian persuasion with multiple receivers. *Available at SSRN 2625399*.
- Zuccarelli, E. (2024, April). Building trust in AI means moving beyond black-box algorithms. Here’s why. <https://www.weforum.org/stories/emerging-technologies/building-trust-in-ai-means-moving-beyond-black-box-algorithms-heres-why/>. Accessed 2026-09-03.

A Proofs of Results

Proposition 1. Suppose $f_S = \beta_0 + \beta_1 x - \delta x^2$ best-fits the data and $\mu_\theta = \mu_\theta(x_\theta, \sigma_\theta)$. Then, for all $f_R \in \phi(f_S; \theta, \mu_\theta)$, we have:

$$f_R(x) = f_S(x) + (\gamma - \delta)(\sigma^2 - (x - x_\theta)^2).$$

Proof. We rewrite the sender's model as a local expansion around x_θ : $f_S(x) = \tilde{\beta}_0 + \tilde{\beta}_1(x - x_\theta) - \delta(x - x_\theta)^2$, where $\tilde{\beta}_1 = \beta_1 - 2\delta x_\theta$. Because sender type intervals θ are non-overlapping, the value of δ uniquely pins down the sender's type θ .

Suppose $f_R \in \phi(f_S; \theta, \mu_\theta)$. Similarly expanding f_R around x_θ , we have:

$$f_R(x) = \tilde{\alpha}_0 + \tilde{\alpha}_1(x - x_\theta) - \gamma(x - x_\theta)^2.$$

Because μ_θ is symmetric, it is elliptical (in one dimension). By Stein's Lemma (Landsman and Nešlehová, 2008), equation (2) can be written as

$$\begin{aligned} \mathbb{E}_{\mu_\theta}[x(f_R(x) - f_S(x))] &= \text{Cov}_{\mu_\theta}(x, f_R(x) - f_S(x)) \\ &= \sigma^2 \mathbb{E}_{\mu_\theta}[f'_R(x) - f'_S(x)] \\ &= \sigma^2 \mathbb{E}_{\mu_\theta}[\tilde{\alpha}_1 - 2\gamma(x - x_\theta) - \tilde{\beta}_1 + 2\delta(x - x_\theta)] = 0 \end{aligned}$$

Because the mean of μ_θ is x_θ , this yields the condition $\tilde{\alpha}_1 = \tilde{\beta}_1$ which is *independent* of σ^2 . This means that $\alpha_1 - 2\gamma x_\theta = \beta_1 - 2\delta x_\theta$, i.e. $\alpha_1 = \beta_1 - 2(\delta - \gamma)x_\theta$. Hence, we have

$$f_R(x) = \alpha_0 + (\beta_1 - 2(\delta - \gamma)x_\theta)x - \gamma x^2,$$

with the value of γ and α_0 potentially indeterminate. However, fixing a plausible γ value, from (10) we have

$$\begin{aligned} \mathbb{E}_{\mu_\theta}[f_R(x)] &= \alpha_0 + (\beta_1 - 2(\delta - \gamma)x_\theta)x_\theta - \gamma(x_\theta^2 + \sigma^2) = \beta_0 + \beta_1 x_\theta - \delta(x_\theta^2 + \sigma^2) = \mathbb{E}_{\mu_\theta}[f_S(x)] \\ &\implies \alpha_0 - 2(\delta - \gamma)x_\theta^2 = \beta_0 - \delta x_\theta^2 - \delta \sigma^2 \\ &\implies \alpha_0 = \beta_0 + (\gamma - \delta)(\sigma^2 - x_\theta^2). \end{aligned}$$

Putting the pieces together, we have

$$\begin{aligned} f_R(x) &= \beta_0 + (\gamma - \delta)(\sigma^2 - x_\theta^2) + (\beta_1 - 2(\delta - \gamma)x_\theta)x - \gamma x^2 \\ &= \beta_0 + \beta_1 x - \delta x^2 + (-(\gamma - \delta)x^2 + (\gamma - \delta)(\sigma^2 - x_\theta^2) + 2(\gamma - \delta)x_\theta x) \\ &= \beta_0 + \beta_1 x - \delta x^2 + (\gamma - \delta)(\sigma^2 - (x - x_\theta)^2) \\ &= f_S(x) + (\gamma - \delta)(\sigma^2 - (x - x_\theta)^2). \end{aligned}$$

□

Theorem 1. *If S estimates naive models, he positions himself as a generalist ($\sigma_\theta = \bar{\sigma}$) and sets $x_\theta = \max\{0, \min\{\mathbb{E}_{f_S}^\theta[x^*(f_S)], \bar{x}\}\}$, where $x^*(f_S)$ is the receiver's choice given f_S . If S estimates cautious models, he positions himself as a specialist ($\sigma_\theta = \underline{\sigma}$) and sets $x_\theta = 0$ or $\bar{x}$.*

Proof. Following from Proposition 1, let $\theta = [\underline{\delta}_\theta, \bar{\delta}_\theta]$. Suppose $\theta^\circ \neq \emptyset$ and $\delta \in \theta^\circ$. Then (3) is necessarily satisfied, and $\phi(f_S; \theta, \mu_\theta) = \{f_S\}$. However, if $\delta = \underline{\delta}_\theta$ or $\bar{\delta}_\theta$, then (3) is a corner solution from the receiver's perspective. If $\delta = \bar{\delta}_\theta$, then any $\gamma \geq \bar{\delta}_\theta$ is plausible from the receiver's perspective. If $\delta = \underline{\delta}_\theta$, any $\gamma \leq \underline{\delta}_\theta$ is plausible. Finally, if $\underline{\delta}_\theta = \bar{\delta}_\theta$, any $\gamma \in [0, \bar{\gamma}]$ value is possible. This pins down the plausible set of γ values from the receiver's perspective.

Next, fix σ . The receiver chooses $x = x^*$ to maximize:

$$\begin{aligned} & \beta_0 + \beta_1 x - \delta x^2 + (\mathbb{E}_\gamma^R[\gamma|f_S] - \delta)(\sigma^2 - (x - x_\theta)^2) \\ & = \beta_0 + \beta_1 x - \delta x^2 + \Delta(f_S)(\sigma^2 - (x - x_\theta)^2) \end{aligned} \quad (12)$$

We note that because the sender behaves as if the receiver has a flat prior over α_0 , her posterior places the same mass on all models that are the same “up to a constant”; hence, changing σ^2 does not change the relative mass she places on each model, fixing a plausible γ value. Thus, by the Envelope Theorem, (12) is increasing in σ^2 if $\Delta(f_S)$ is positive and decreasing if $\Delta(f_S)$ is negative. Working backwards, when the sender takes expectations over (12), he chooses $\sigma = \bar{\sigma}$ if $\mathbb{E}_{f_S}[\Delta(f_S)] \geq 0$ and $\underline{\sigma}$ otherwise.

Finally, we characterize the equilibrium value of x_θ . Note that because the sender behaves as if the receiver's prior is flat over α_0 and α_1 , his expectation of the receiver's expectation of γ does not depend on x_θ . Returning to (12) and using the envelope theorem, we have that the derivative with respect to x_θ is:

$$2\Delta(f_S)(x^* - x_\theta)$$

This is (strictly) concave if $\Delta(f_S) > 0$, and achieves a maximum when $x^* = x_\theta$. If $\Delta(f_S) \leq 0$, the function is convex and achieves a maximum at a corner solution 0 or $\bar{x}$, giving the potential values for x_θ . Given S must commit to data collection beforehand and integrating backwards, this gives $x_\theta = \mathbb{E}_{f_S}^\theta[x^*]$ if he expects $\Delta(f_S) > 0$ and $x_\theta = 0$ or $\bar{x}$ otherwise.

Finally, we note that $x^*(f_S)$ solves

$$\begin{aligned} \beta_1 - 2\delta x^* - 2\Delta(f_S)(x^* - x_\theta) &= 0 \implies x^* = \frac{\beta_1 + 2\Delta(f_S)x_\theta}{2\mathbb{E}_\gamma^R[\gamma|f_S, \theta]} \\ \implies \mathbb{E}_{f_S}^\theta[x^*(f_S)] &= \mathbb{E}_{1,\delta}^\theta \left[\frac{\beta_1 + 2\Delta(f_S)x_\theta}{2\delta + 2\Delta(f_S)} \right]. \end{aligned}$$

□

Proposition 3. *Suppose $f_S = \beta_0 + \beta_1'x - x'\delta x$ best-fits the data and $\mu_\theta = \mu_\theta(x_\theta, \Sigma_\theta)$. Then, for all*

$f_R \in \phi(f_S; \theta, \mu_\theta)$, we have:

$$f_R(x) = f_S(x) + \text{tr}((\gamma - \delta)\Sigma_\theta) - (x - x_\theta)'(\gamma - \delta)(x - x_\theta).$$

Proof. As above, we necessarily have the following moment conditions for all i :

$$\mathbb{E}_{\mu_\theta}[x_i(f_R(x) - f_S(x))] = 0 \quad \mathbb{E}_{\mu_\theta}[f_R(x) - f_S(x)] = 0$$

and we may or may not have the correct moment condition with respect to the entries of γ . Crucially, however, because μ_θ is elliptical by Stein's Lemma for elliptical random vectors (Landsman and Nešlehová, 2008), we can simplify the first equation to

$$\begin{aligned} \mathbb{E}_{\mu_\theta}[x_i(f_R(x) - f_S(x))] = 0 &\implies \mathbb{E}_{\mu_\theta}\left[\frac{\partial}{\partial x_i} f_R(x)\right] = \mathbb{E}_{\mu_\theta}\left[\frac{\partial}{\partial x_i} f_S(x)\right] \quad \forall i \\ &\implies \alpha - (\gamma + \gamma')\mathbb{E}_{\mu_\theta}[x] = \beta - (\delta + \delta')\mathbb{E}_{\mu_\theta}[x] \\ &\implies \alpha = \beta + ((\gamma + \gamma') - (\delta + \delta'))x_\theta \\ &\implies \alpha'x = \beta'x + x_\theta'((\gamma + \gamma') - (\delta + \delta'))x \end{aligned}$$

which is invariant to Σ_θ . Next, working with the second equation, we have

$$\mathbb{E}_{\mu_\theta}[f_R(x)] = \alpha_0 + \alpha x_\theta - (x_\theta + \Sigma_\theta^{\frac{1}{2}}\epsilon)' \gamma (x_\theta + \Sigma_\theta^{\frac{1}{2}}\epsilon) = \beta_0 + \beta x_\theta - (x_\theta + \Sigma_\theta^{\frac{1}{2}}\epsilon)' \delta (x_\theta + \Sigma_\theta^{\frac{1}{2}}\epsilon) = \mathbb{E}_{\mu_\theta}[f_S(x)].$$

Note that

$$(x_\theta + \Sigma_\theta^{\frac{1}{2}}\epsilon)' \gamma (x_\theta + \Sigma_\theta^{\frac{1}{2}}\epsilon) = x_\theta' \gamma x_\theta + 2x_\theta' \gamma \Sigma_\theta^{\frac{1}{2}}\epsilon + \Sigma_\theta^{\frac{1}{2}}\epsilon' \gamma \Sigma_\theta^{\frac{1}{2}}\epsilon$$

and taking the expectation with respect to ϵ thus gives

$$x_\theta' \gamma x_\theta + \text{tr}(\gamma \Sigma_\theta).$$

Utilizing the same logic with δ and plugging in the α equation gives

$$\begin{aligned} \alpha_0 &= \beta_0 + (\beta - \alpha)x_\theta + x_\theta'(\gamma - \delta)x_\theta + \text{tr}((\gamma - \delta)\Sigma_\theta) \\ &= \beta_0 - x_\theta'((\gamma + \gamma') - (\delta + \delta'))x_\theta + \text{tr}((\gamma - \delta)\Sigma_\theta) \\ &= \beta_0 - x_\theta'(\gamma - \delta)x_\theta + \text{tr}((\gamma - \delta)\Sigma_\theta) \end{aligned}$$

Hence, we renormalize f_R , plug in the values of α_0 , α , and reorganize the quadratic terms to get:

$$\begin{aligned} f_R(x) &= \alpha_0 + \alpha'x - x'\gamma x \\ &= \beta_0 - x_\theta'(\gamma - \delta)x_\theta + \text{tr}((\gamma - \delta)\Sigma_\theta) + \beta'x + x_\theta'((\gamma + \gamma') - (\delta + \delta'))x - x'\gamma x \\ &= \beta_0 + \beta'x + \text{tr}((\gamma - \delta)\Sigma_\theta) - (x - x_\theta)'(\gamma - \delta)(x - x_\theta). \end{aligned}$$

□

Theorem 2. *Senders who estimate naive models in x_i position themselves as generalists. Those who estimate cautious models in x_i position themselves as specialists. I.e.,*

- if $\mathbb{E}_\delta^\theta[\Delta_{ii}(\delta)] > 0$, then $\sigma_{i,\theta} = \bar{\sigma}$.
- If $\mathbb{E}_\delta^\theta[\Delta_{ii}(\delta)] < 0$, then $\sigma_{i,\theta} = \underline{\sigma}$.

Moreover, senders who neglect interactions between x_i and x_j create collinearity between x_i and x_j . Senders who exaggerate interactions create anticollinearity between x_i and x_j . I.e.,

- if $\mathbb{E}_\delta^\theta[\Delta_{ij}(\delta) + \Delta_{ji}(\delta)] > 0$, then $\Sigma_{ij,\theta} = \bar{\rho}\sigma_{i,\theta}\sigma_{j,\theta}$.
- If $\mathbb{E}_\delta^\theta[\Delta_{ij}(\delta) + \Delta_{ji}(\delta)] < 0$, then $\Sigma_{ij,\theta} = -\bar{\rho}\sigma_{i,\theta}\sigma_{j,\theta}$.

Proof. The proof of this result follows nearly identically to that of Theorem 1. The key is that

$$\begin{aligned}\mathbb{E}_\gamma^R[\text{tr}((\gamma - \delta)\Sigma_\theta)|\delta, \theta] &= \mathbb{E}_\gamma^R\left[\sum_{i=1}^n(\gamma_{ii} - \delta_{ii})\Sigma_{ii,\theta} + \sum_{i < j}[(\gamma_{ij} + \gamma_{ji}) - (\delta_{ij} + \delta_{ji})]\Sigma_{ij,\theta}|\delta, \theta\right] \\ &= \mathbb{E}_\gamma^R\left[\sum_{i=1}^n\Delta_{ii}(\delta)\Sigma_{ii,\theta} + \sum_{i < j}[\Delta_{ij}(\delta) + \Delta_{ji}(\delta)]\Sigma_{ij,\theta}|\delta, \theta\right]\end{aligned}$$

which generates corner solutions depending on the expected signs of Δ_{ii} and $\Delta_{ij} + \Delta_{ji}$. □

Proposition 4. *Fix θ . The value of information $V(\theta)$ for the receiver from hiring a type θ sender is given by*

$$V(\theta) = \mathbb{E}_{f_R}^R\left[\mathbb{E}_\gamma^R[\gamma|f'_R(x_\theta), \delta_\theta(\gamma)]\left(x^* - \frac{\mathbb{E}_1^R[\alpha_1]}{2\mathbb{E}_\gamma^R[\gamma]}\right)^2\right],$$

where $\delta_\theta(\gamma) := \min\{\max\{\gamma, \underline{\delta}_\theta\}, \bar{\delta}_\theta\}$ and

$$x^* = \max\left\{\underline{x}, \min\left\{\bar{x}, x_\theta + \frac{f'_R(x_\theta)}{2\mathbb{E}_\gamma^R[\gamma|f'_R(x_\theta), \delta_\theta(\gamma)]}\right\}\right\}$$

Proof. Suppose x^* maximizes $\mathbb{E}_{f_R|f_S,\theta}^R[f_R(x)|f_S(x)]$ and x_0 maximizes $\mathbb{E}_{f_R}^R[f_R(x)]$. Conditional

on f_S , we have:

$$\begin{aligned}
& \mathbb{E}_{f_R|f_S,\theta}^R[f_R(x^*)] - \mathbb{E}_{f_R|f_S,\theta}^R[f_R(x_0)] \\
&= f_S(x^*) + \Delta(f_S)(\sigma_\theta^2 - (x^* - x_\theta)^2) - f_S(x_0) - \Delta(f_S)(\sigma_\theta^2 - (x_0 - x_\theta)^2) \\
&= \beta_1(x^* - x_0) - \delta(x^{*2} - x_0^2) + \Delta(f_S)((x_0 - x_\theta)^2 - (x^* - x_\theta)^2) \\
&= \beta_1(x^* - x_0) - \delta(x^{*2} - x_0^2) + \Delta(f_S)(2x_\theta(x^* - x_0) - (x^{*2} - x_0^2)) \\
&= (\beta_1 + 2\Delta(f_S)x_\theta)(x^* - x_0) - (\delta + \Delta(f_S))(x^{*2} - x_0^2) \\
&= (x^* - x_0)[\beta_1 + 2\Delta(f_S)x_\theta - (\delta + \Delta(f_S))(x^* + x_0)] \\
&= (x^* - x_0)[2(\delta + \Delta(f_S))x^* - (\delta + \Delta(f_S))(x^* + x_0)] \\
&= (\delta + \Delta(f_S))(x^* - x_0)^2 \\
&= \mathbb{E}_\gamma^R[\gamma|\theta, f_S](x^*(f_S) - x_0)^2.
\end{aligned}$$

In turn, we have

$$V(\theta) = \mathbb{E}_{f_S|\theta}^R[\mathbb{E}_\gamma^R[\gamma|\theta, f_S](x^*(f_S) - x_0)^2].$$

In particular, we have $x_0 = \frac{\mathbb{E}_1^R[\alpha_1]}{2\mathbb{E}_\gamma^R[\gamma]}$ and, from the previous result,

$$x^* = \frac{\beta_1 + 2\Delta(f_S)x_\theta}{2\mathbb{E}_\gamma^R[\gamma|f_S, \theta]} = x_\theta + \frac{\beta_1 - 2\delta x_\theta}{2\mathbb{E}_\gamma^R[\gamma|f_S, \theta]}.$$

This leaves us with:

$$V(\theta) = \mathbb{E}_{f_S|\theta}^R\left[\mathbb{E}_\gamma^R[\gamma|\theta, f_S]\left(x_\theta + \frac{\beta_1 - 2\delta x_\theta}{2\mathbb{E}_\gamma^R[\gamma|f_S, \theta]} - \frac{\mathbb{E}_1^R[\alpha_1]}{2\mathbb{E}_\gamma^R[\gamma]}\right)^2\right].$$

Note moreover that $\beta_1 - 2\delta x_\theta = f'_S(x_\theta) = f'_R(x_\theta)$. This means that we can write the value as:

$$V(\theta) = \mathbb{E}_{f_R}^R\left[\mathbb{E}_\gamma^R[\gamma|f'_R(x_\theta), \delta_\theta(\gamma)]\left(x_\theta + \frac{f'_R(x_\theta)}{2\mathbb{E}_\gamma^R[\gamma|f'_R(x_\theta), \delta_\theta(\gamma)]} - \frac{\mathbb{E}_1^R[\alpha_1]}{2\mathbb{E}_\gamma^R[\gamma]}\right)^2\right],$$

where $\delta_\theta(\gamma) := \min\{\max\{\gamma, \underline{\delta}_\theta\}, \bar{\delta}_\theta\}$. □

Proposition 5. Writing the value of information $V(\theta)$ as a function of the parameter x_θ , $V(\theta, x_\theta)$, we have

$$V_{x_\theta}(\theta, x_\theta) = -\mathbb{E}_{f_R}^R\left[\left(\frac{\partial}{\partial x_\theta}\mathbb{E}_\gamma^R[\gamma|f'_R(x_\theta), \delta_\theta(\gamma)]\right)\left(\frac{f'_R(x_\theta)}{2\mathbb{E}_\gamma^R[\gamma|f'_R(x_\theta), \delta_\theta(\gamma)]}\right)^2\right].$$

Moreover if p_1^R is absolutely continuous as defined on all of $\mathbb{R}$ and $\lim_{|\alpha_1| \rightarrow \infty} \alpha_1^2 p_1^R(\alpha_1) = 0$, $\arg \max_{x_\theta} V(\theta, x_\theta) >$

0.

Proof. Recalling that $f_R''(x_\theta) = -2\gamma$ and using the previous proposition, we note that

$$\begin{aligned} \frac{\partial V(\theta)}{\partial x_\theta} = & \mathbb{E}_{f_R}^R \left[\left(\frac{\partial}{\partial x_\theta} \mathbb{E}_\gamma^R[\gamma | f_R'(x_\theta), \delta_\theta(\gamma)] \right) \left(x_\theta + \frac{f_R'(x_\theta)}{2\mathbb{E}_\gamma^R[\gamma | f_R'(x_\theta), \delta_\theta(\gamma)]} - \frac{\mathbb{E}_1^R[\alpha_1]}{2\mathbb{E}_\gamma^R[\gamma]} \right)^2 \right. \\ & + 2\mathbb{E}_\gamma^R[\gamma | f_R'(x_\theta), \delta_\theta(\gamma)] \left(x_\theta + \frac{f_R'(x_\theta)}{2\mathbb{E}_\gamma^R[\gamma | f_R'(x_\theta), \delta_\theta(\gamma)]} - \frac{\mathbb{E}_1^R[\alpha_1]}{2\mathbb{E}_\gamma^R[\gamma]} \right) \\ & \cdot \left(1 - \frac{\gamma}{\mathbb{E}_\gamma^R[\gamma | f_R'(x_\theta), \delta_\theta(\gamma)]} - \frac{f_R'(x_\theta)}{2\mathbb{E}_\gamma^R[\gamma | f_R'(x_\theta), \delta_\theta(\gamma)]^2} \frac{\partial}{\partial x_\theta} \mathbb{E}_\gamma^R[\gamma | f_R'(x_\theta), \delta_\theta(\gamma)] \right) \Bigg] \end{aligned}$$

In particular, we note that by Bayes' Rule:

$$\mathbb{E}_{f_R}^R [\mathbb{E}_\gamma^R[\gamma | f_R'(x_\theta), \delta_\theta(\gamma)]] = \mathbb{E}_\gamma^R[\gamma] \implies \mathbb{E}_{f_R}^R \left[\frac{\partial}{\partial x_\theta} \mathbb{E}_\gamma^R[\gamma | f_R'(x_\theta), \delta_\theta(\gamma)] \right] = 0.$$

Using this we simplify the above expression to

$$\begin{aligned} & \mathbb{E}_{f_R}^R \left[\left(\frac{\partial}{\partial x_\theta} \mathbb{E}_\gamma^R[\gamma | f_R'(x_\theta), \delta_\theta(\gamma)] \right) \left(x_\theta + \frac{f_R'(x_\theta)}{2\mathbb{E}_\gamma^R[\gamma | f_R'(x_\theta), \delta_\theta(\gamma)]} - \frac{\mathbb{E}_1^R[\alpha_1]}{2\mathbb{E}_\gamma^R[\gamma]} \right) \right. \\ & \cdot \left(x_\theta + \frac{f_R'(x_\theta)}{2\mathbb{E}_\gamma^R[\gamma | f_R'(x_\theta), \delta_\theta(\gamma)]} - \frac{\mathbb{E}_1^R[\alpha_1]}{2\mathbb{E}_\gamma^R[\gamma]} - \frac{f_R'(x_\theta)}{\mathbb{E}_\gamma^R[\gamma | f_R'(x_\theta), \delta_\theta(\gamma)]} \right) \\ & \left. + 2 \left(x_\theta + \frac{f_R'(x_\theta)}{2\mathbb{E}_\gamma^R[\gamma | f_R'(x_\theta), \delta_\theta(\gamma)]} - \frac{\mathbb{E}_1^R[\alpha_1]}{2\mathbb{E}_\gamma^R[\gamma]} \right) \cdot \left(\mathbb{E}_\gamma^R[\gamma | f_R'(x_\theta), \delta_\theta(\gamma)] - \gamma \right) \right] \\ & \quad \quad \quad = 0 \text{ conditioning on } f_R' \\ & = \mathbb{E}_{f_R}^R \left[\left(\frac{\partial}{\partial x_\theta} \mathbb{E}_\gamma^R[\gamma | f_R'(x_\theta), \delta_\theta(\gamma)] \right) \left(x_\theta + \frac{f_R'(x_\theta)}{2\mathbb{E}_\gamma^R[\gamma | f_R'(x_\theta), \delta_\theta(\gamma)]} - \frac{\mathbb{E}_1^R[\alpha_1]}{2\mathbb{E}_\gamma^R[\gamma]} \right) \left(x_\theta - \frac{f_R'(x_\theta)}{2\mathbb{E}_\gamma^R[\gamma | f_R'(x_\theta), \delta_\theta(\gamma)]} - \frac{\mathbb{E}_1^R[\alpha_1]}{2\mathbb{E}_\gamma^R[\gamma]} \right) \right] \\ & = \mathbb{E}_{f_R}^R \left[\left(\frac{\partial}{\partial x_\theta} \mathbb{E}_\gamma^R[\gamma | f_R'(x_\theta), \delta_\theta(\gamma)] \right) \left(\left(x_\theta - \frac{\mathbb{E}_1^R[\alpha_1]}{2\mathbb{E}_\gamma^R[\gamma]} \right)^2 - \left(\frac{f_R'(x_\theta)}{2\mathbb{E}_\gamma^R[\gamma | f_R'(x_\theta), \delta_\theta(\gamma)]} \right)^2 \right) \right] \\ & = -\mathbb{E}_{f_R}^R \left[\left(\frac{\partial}{\partial x_\theta} \mathbb{E}_\gamma^R[\gamma | f_R'(x_\theta), \delta_\theta(\gamma)] \right) \left(\frac{f_R'(x_\theta)}{2\mathbb{E}_\gamma^R[\gamma | f_R'(x_\theta), \delta_\theta(\gamma)]} \right)^2 \right]. \end{aligned}$$

Finally, we evaluate this expression at $x_\theta = 0$. In this case, note that $f_R'(0) = \alpha_1$, so the expression simplifies to

$$-\mathbb{E}_{f_R}^R \left[\left(\frac{\partial}{\partial x_\theta} \Big|_{x_\theta=0} \mathbb{E}_\gamma^R[\gamma | f_R'(x_\theta), \delta_\theta(\gamma)] \right) \left(\frac{\alpha_1}{2\mathbb{E}_\gamma^R[\gamma | \delta_\theta(\gamma)]} \right)^2 \right].$$

Next, we express R 's posterior mean of γ given $f'_R(x_\theta)$ and $\delta_\theta(\gamma)$ as

$$\mathbb{E}_\gamma^R[\gamma|f'_R(x_\theta), \delta_\theta(\gamma)] = \frac{\mathbb{E}_{\tilde{\gamma}}^R\left[\frac{p_1^R(\alpha_1 - 2\gamma x_\theta + 2\tilde{\gamma} x_\theta)}{p_1^R(\alpha_1)} \tilde{\gamma} | \delta_\theta(\gamma)\right]}{\mathbb{E}_{\tilde{\gamma}}^R\left[\frac{p_1^R(\alpha_1 - 2\gamma x_\theta + 2\tilde{\gamma} x_\theta)}{p_1^R(\alpha_1)} | \delta_\theta(\gamma)\right]}$$

Differentiating with respect to x_θ yields

$$\frac{\mathbb{E}_{\tilde{\gamma}}^R\left[\frac{2(\tilde{\gamma} - \gamma)p_1^{R'}(\alpha_1 - 2\gamma x_\theta + 2\tilde{\gamma} x_\theta)}{p_1^R(\alpha_1)} \tilde{\gamma} | \delta_\theta(\gamma)\right]}{\mathbb{E}_{\tilde{\gamma}}^R\left[\frac{p_1^R(\alpha_1 - 2\gamma x_\theta + 2\tilde{\gamma} x_\theta)}{p_1^R(\alpha_1)} | \delta_\theta(\gamma)\right]} - \frac{\mathbb{E}_{\tilde{\gamma}}^R\left[\frac{2(\tilde{\gamma} - \gamma)p_1^{R'}(\alpha_1 - 2\gamma x_\theta + 2\tilde{\gamma} x_\theta)}{p_1^R(\alpha_1)} | \delta_\theta(\gamma)\right] \mathbb{E}_{\tilde{\gamma}}^R\left[\frac{p_1^R(\alpha_1 - 2\gamma x_\theta + 2\tilde{\gamma} x_\theta)}{p_1^R(\alpha_1)} \tilde{\gamma} | \delta_\theta(\gamma)\right]}{\mathbb{E}_{\tilde{\gamma}}^R\left[\frac{p_1^R(\alpha_1 - 2\gamma x_\theta + 2\tilde{\gamma} x_\theta)}{p_1^R(\alpha_1)} | \delta_\theta(\gamma)\right]^2}.$$

But at $x_\theta = 0$ this can be heavily simplified to

$$\begin{aligned} & \mathbb{E}_{\tilde{\gamma}}^R\left[\frac{2(\tilde{\gamma} - \gamma)p_1^{R'}(\alpha_1)}{p_1^R(\alpha_1)} \tilde{\gamma} | \delta_\theta(\gamma)\right] - \mathbb{E}_{\tilde{\gamma}}^R\left[\frac{2(\tilde{\gamma} - \gamma)p_1^{R'}(\alpha_1)}{p_1^R(\alpha_1)} | \delta_\theta(\gamma)\right] \mathbb{E}_{\tilde{\gamma}}^R[\tilde{\gamma} | \delta_\theta(\gamma)] \\ &= \frac{2p_1^{R'}(\alpha_1)}{p_1^R(\alpha_1)} \left(\mathbb{E}_{\tilde{\gamma}}^R[\tilde{\gamma}(\tilde{\gamma} - \gamma) | \delta_\theta(\gamma)] - \mathbb{E}_{\tilde{\gamma}}^R[(\tilde{\gamma} - \gamma) | \delta_\theta(\gamma)] \mathbb{E}_{\tilde{\gamma}}^R[\tilde{\gamma} | \delta_\theta(\gamma)] \right) \\ &= \frac{2p_1^{R'}(\alpha_1)}{p_1^R(\alpha_1)} \text{Var}_{\tilde{\gamma}}^R(\tilde{\gamma} | \delta_\theta(\gamma)) \end{aligned}$$

Plugging this into the original equation yields

$$\begin{aligned} \left. \frac{\partial V(\theta)}{\partial x_\theta} \right|_{x_\theta=0} &= -\mathbb{E}_{f_R}^R \left[\frac{2p_1^{R'}(\alpha_1)}{p_1^R(\alpha_1)} \text{Var}_{\tilde{\gamma}}^R(\tilde{\gamma} | \delta_\theta(\gamma)) \left(\frac{\alpha_1}{2\mathbb{E}_{\tilde{\gamma}}^R[\gamma | \delta_\theta(\gamma)]} \right)^2 \right] \\ &= -2\mathbb{E}_{f_R}^R \left[\frac{p_1^{R'}(\alpha_1)}{p_1^R(\alpha_1)} \alpha_1^2 \right] \mathbb{E}_{f_R}^R \left[\frac{\text{Var}_{\tilde{\gamma}}^R(\tilde{\gamma} | \delta_\theta(\gamma))}{\mathbb{E}_{\tilde{\gamma}}^R[\gamma | \delta_\theta(\gamma)]^2} \right] \end{aligned}$$

But note that using integration by parts, the first term can be expressed as:

$$\mathbb{E}_{f_R}^R \left[\frac{p_1^{R'}(\alpha_1)}{p_1^R(\alpha_1)} \alpha_1^2 \right] = \int_{\mathbb{R}} \alpha_1^2 p_1^{R'}(\alpha_1) d\alpha_1 = \alpha_1^2 p_1^R(\alpha_1) \Big|_{-\infty}^{+\infty} - \int_{\mathbb{R}} 2\alpha_1 p_1^R(\alpha_1) d\alpha_1 = -2\mathbb{E}_1^R[\alpha_1].$$

This shows that the derivative of $V(\theta)$ is *positive* at $x_\theta = 0$, and hence that its maximizer is strictly positive. $\square$

Proposition 6. Fix a sender type θ and let $x^*(f_S)$ maximize $\mathbb{E}_{f_R|f_S, \theta, \mu_\theta}[y|x]$. Then, there exists $\sigma_\theta^* \in (\underline{\sigma}, \overline{\sigma})$ that maximizes

$$W(\sigma_\theta) := \mathbb{E}_{f_S|\theta}^R[\text{Var}_{f_R|f_S}^R(y|x^*(f_S), x_\theta, \sigma_\theta)].$$

with respect to σ_θ . Moreover, W is strictly decreasing in σ for $\sigma < \sigma_\theta^*$ and increasing for $\sigma > \sigma_\theta^*$.

Proof. With some abuse of notation, given f_S and the other parameters, we have:

$$\begin{aligned}\text{Var}(y|x^*) &= \text{Var}(f_S(x^*(f_S)) + (\gamma - \delta)(\sigma_\theta^2 - (x^*(f_S) - x_\theta)^2)) \\ &= ((\sigma_\theta^2 - (x^*(f_S) - x_\theta)^2))^2 \text{Var}(\gamma|f_S, \theta, x_\theta),\end{aligned}$$

where we note that information about γ depends on both f_S as well as x_θ . If $\theta^\circ \neq \emptyset$ and $\delta \in \theta^\circ$, then this variance is 0. Otherwise, if $\gamma \in \partial\theta$ (the boundary of θ), then the variance is > 0 . Hence, our problem is to choose x_θ and σ_θ to minimize

$$\mathbb{E}_{f_S|\theta}^R [((\sigma_\theta^2 - (x^*(f_S) - x_\theta)^2))^2 \text{Var}(\gamma|f_S, \theta, x_\theta)].$$

We note that this function is quadratic in σ_θ^2 and minimized at

$$\sigma_\theta^2 = \frac{\mathbb{E}_{f_S|\theta}^R [(x^*(f_S) - x_\theta)^2 \text{Var}(\gamma|f_S, \theta, x_\theta)]}{\mathbb{E}_{f_S|\theta}^R [\text{Var}(\gamma|f_S, \theta, x_\theta)]} = \frac{\mathbb{E}_{f_S|\theta}^R [(\frac{\beta_1 - 2\delta x_\theta}{2\mathbb{E}_{f_S|\theta}^R [\gamma|f_S, \theta]})^2 \text{Var}(\gamma|f_S, \theta, x_\theta)]}{\mathbb{E}_{f_S|\theta}^R [\text{Var}(\gamma|f_S, \theta, x_\theta)]}$$

The terms in the numerator represent the ex-ante expectation of the squared difference between x_θ and the eventual choice $x^*(f_S)$ multiplied by the variance of γ . The term in the denominator is the expected variance of γ . Because there is uncertainty about $x^*(f_S)$, the numerator is always positive and bounded away from 0, which gives the result. □

Proposition 7. Suppose R chooses $x^*(f_S)$ to maximize (5). Then:

1. A naive sender chooses $\sigma_\theta > \sigma_\theta^*$ and a cautious sender chooses $\sigma_\theta < \sigma_\theta^*$
2. As $\psi \rightarrow 1$, $x^*(f_S) \rightarrow x_\theta \pm \sigma$.

Proof. Given f_S and θ and using the previous result, R 's problem is to maximize

$$\max_x (1 - \psi)(f_S(x) + \Delta(f_S)(\sigma_\theta^2 - (x - x_\theta)^2)) - \psi((\sigma_\theta^2 - (x - x_\theta)^2))^2 \text{Var}(\gamma|f_S),$$

which S then integrates out over the plausible values of f_S . Denote this maximizer by $x^*(f_S)$. By the envelope theorem, the derivative with respect to σ_θ^2 is

$$\mathbb{E}_{f_S} [(1 - \psi)\Delta(f_S) - 2\psi(\sigma_\theta^2 - (x^*(f_S) - x_\theta)^2)\text{Var}(\gamma|f_S)].$$

Recall from above that the expectation of $(\sigma_\theta^2 - (x^*(f_S) - x_\theta)^2)\text{Var}(\gamma|f_S)$ is positive below σ_θ^* and negative above. For a naive sender, $\Delta(f_S) > 0$, meaning the derivative is still positive at σ_θ^* , and thus that his choice of σ is higher. The opposite holds for the cautious sender.

The second result follows by noting that as $\psi \rightarrow 1$, R is simply choosing x^* to minimize variance. But this can conditionally always be achieved by setting $x^* = x_\theta \pm \sigma_\theta$, which gives the result. □

Theorem 3. Suppose Assumption 9 holds. If S estimates naive models, he positions himself as a generalist ($\sigma_\theta = \bar{\sigma}$) and sets $x_\theta = \max\{0, \min\{\frac{\mathbb{E}_1^\theta[\beta_1]}{2\mathbb{E}_\delta^\theta[\delta]}, \bar{x}\}\}$. If S estimates cautious models, he positions himself as a specialist ($\sigma_\theta = \underline{\sigma}$) and sets $x_\theta = 0$ or $\bar{x}$.

Alternatively, suppose Assumption 10 holds, suppose $\bar{\sigma}$ is sufficiently large, and fix x_θ . If S estimates naive models, he strictly prefers $\sigma_\theta = \bar{\sigma}$ to $\underline{\sigma}$. If S estimates cautious models, he strictly prefers $\sigma_\theta = \underline{\sigma}$ to $\bar{\sigma}$.

We prove this result via the following lemmata. The first part of the theorem (assuming Assumption 9) follows immediately from the first lemma after applying the same reasoning as in Theorem 1.

Lemma 1. Suppose Assumption 9 holds and fix f_S . Then, for all $f_R \in \phi(f_S; \theta, \mu_\theta)$, there exists $L(\sigma)$ strictly increasing in σ with $L(0) = 0$ such that:

$$f_R(x) = f_S(x) + (\gamma - \delta)[L(\sigma) - \ell(x - x_\theta)].$$

Proof. Suppose $f_R \in \phi(f_S; \theta, \mu_\theta)$ for $f_S = \beta_0 + \beta_1 x - \delta \ell(x - x_\theta)$. This means that we have

$$f_R(x) = \tilde{\alpha}_0 + \tilde{\alpha}_1 x - \gamma \ell(x - x_\theta).$$

Applying Stein's Lemma as above, the moment condition with respect to $\tilde{\alpha}_1$ can be written as

$$\begin{aligned} \mathbb{E}_{\mu_\theta}[x(f_R(x) - f_S(x))] &= \text{Cov}_{\mu_\theta}(x, f_R(x) - f_S(x)) \\ &= \sigma^2 \mathbb{E}_{\mu_\theta}[f'_R(x) - f'_S(x)] \\ &= \sigma^2 \mathbb{E}_{\mu_\theta}[\tilde{\alpha}_1 - \gamma \ell'(x - x_\theta) - \beta_1 + \delta \ell'(x - x_\theta)] = 0. \end{aligned}$$

Because $\ell(x - x_\theta)$ is symmetric around x_θ , the terms with ℓ' are equal to 0. Thus, we have $\tilde{\alpha}_1 = \beta_1$ for all σ and are left to solve for $\tilde{\alpha}_0$. But for each γ value, as above, we have

$$\begin{aligned} \mathbb{E}_{\mu_\theta}[f_R(x)] = \mathbb{E}_{\mu_\theta}[f_S(x)] &\implies \tilde{\alpha}_0 - \gamma \mathbb{E}_{\mu_\theta, \sigma}[\ell(x - x_\theta)] = \beta_0 - \delta \mathbb{E}_{\mu_\theta, \sigma}[\ell(x - x_\theta)] \\ &\implies \tilde{\alpha}_0 = \beta_0 + (\gamma - \delta) \mathbb{E}_{\mu_\theta, \sigma}[\ell(x - x_\theta)]. \end{aligned}$$

Defining $L(\sigma) = \mathbb{E}_{\mu_\theta, \sigma}[\ell(x - x_\theta)]$, we note that because $\ell(x - x_\theta)$ is convex, $L(\sigma)$ is increasing in σ . As $\sigma \rightarrow 0$, $L(\sigma) \rightarrow 0$. This gives the result. $\square$

Lemma 2. Suppose Assumption 10 holds and fix f_S . Then, for each $f_R \in \phi(f_S; \theta, \mu_\theta)$, $f_R(x_\theta)$ is increasing (decreasing).

Proof. As above, we write $f_R(x) = \tilde{\alpha}_0 + \tilde{\alpha}_1 x - \gamma m(x)$. Applying Stein's Lemma again to the slope moment condition, we have $\mathbb{E}_{\mu_\theta}[f'_R(x)] = \mathbb{E}_{\mu_\theta}[f'_S(x)]$. Using this condition, we implicitly define $\alpha_1(\sigma)$ as

$$\alpha_1(\sigma) = \beta_1 + (\gamma - \delta) \mathbb{E}_{\mu_\theta}[m'(x)]$$

and note that its sign with respect to σ is ambiguous. Similarly, we define $\alpha_0(\sigma)$ implicitly as

$$\alpha_0(\sigma) = \beta_0 + (\beta_1 - \alpha_1(\sigma))x_\theta + (\gamma - \delta)\mathbb{E}_{\mu_\theta, \sigma}[m(x)]$$

and note in particular that, by definition, we have

$$\mathbb{E}_{\mu_\theta, \sigma}[f_S(x)] = \mathbb{E}_{\mu_\theta, \sigma}[f_R(x)] = \mathbb{E}_{\mu_\theta, \sigma}[\alpha_0(\sigma)x + \alpha_1(\sigma) - \gamma m(x)]$$

But then, by the envelope theorem, we have that

$$\begin{aligned} \frac{\partial}{\partial \sigma} f_R(x_\theta | \sigma) &= \alpha'_0(\sigma) + \alpha'_1(\sigma)x_\theta - \gamma m(x_\theta) \\ &= -\alpha'_1(\sigma)x_\theta + (\gamma - \delta)\frac{\partial}{\partial \sigma} \mathbb{E}_{\mu_\theta, \sigma}[m(x)] + \alpha'_1(\sigma)x_\theta \\ &= (\gamma - \delta)\frac{\partial}{\partial \sigma} \mathbb{E}_{\mu_\theta, \sigma}[m(x)] \end{aligned}$$

Since m is convex, this is weakly positive if and only if $\gamma \geq \delta$. Moreover, as $\sigma \rightarrow \infty$, $\mathbb{E}_{\mu_\theta, \sigma}[m(x)]$ is bounded away from 0, meaning that $f_R(x_\theta)$ approaches ∞ as $\sigma \rightarrow \infty$.

We note that because the sender behaves as if the receiver has a flat prior over α_0 and α_1 , her posterior places the same mass on all models that are the same “up to a constant and linear term.” This means that for $\gamma > \delta$, there exists σ such that $f_R(x_\theta | \sigma) > f_R(x^* | \underline{\sigma})$, where x^* maximizes $f_R(x | \underline{\sigma})$. Hence, S is better off being a generalist than a specialist when $\gamma > \delta$. The same logic shows that S prefers being a specialist when $\gamma < \delta$. Applying the same logic as in the proof of Theorem 1 gives the result. $\square$

Proposition 8. *Consider two senders θ, θ' such that $\theta \cap \theta' \neq \emptyset$. Suppose R receives information from either sender with positive probability. Then, a type θ sender’s utility either strictly increases or decreases only if $\partial\theta' \cap \theta \neq \emptyset$ or $\partial\theta \cap \theta' \neq \emptyset$.*

Proof. We show the following cases exhaustively. Given $f_S(x)$, a type θ sender’s utility strictly increases (decreases) from the existence of type θ' if one of the following holds.

1. $\delta \in \theta^\circ \neq \emptyset$, $\delta = \bar{\delta}'$, and θ' distantly (locally) samples.
2. $\delta \in \theta^\circ \neq \emptyset$, $\delta = \underline{\delta}'$, and θ' locally (distantly) samples.
3. $\delta = \bar{\delta}$, $\theta^\circ = \emptyset$, type θ locally samples, and θ' distantly (locally) samples.
4. $\delta = \underline{\delta}$, $\theta^\circ = \emptyset$, type θ distantly samples, and θ' locally (distantly) samples.
5. $\theta^\circ = \emptyset$, $\theta'^\circ \neq \emptyset$, $\delta = \bar{\delta}'$, and θ' distantly (locally) samples.
6. $\theta^\circ = \emptyset$, $\theta'^\circ \neq \emptyset$, $\delta = \underline{\delta}'$, and θ' locally (distantly) samples.

First, we show how the receiver's expected value from entertaining models that have steeper or shallower tradeoffs than the sender's affects her maximal expected utility. Suppose $\gamma > \delta$. Given a sampling strategy σ , we have:

$$\begin{aligned} \max \beta_0 + \beta_1 x - \delta(x - x_\theta)^2 &= \beta_0 + \beta_1 x_\theta + \frac{\beta_1^2}{4\delta} \\ \max \beta_0 + \beta_1 x - \gamma(x - x_\theta)^2 + (\gamma - \delta)\sigma^2 &= \beta_0 + \beta_1 x_\theta + \frac{\beta_1^2}{4\gamma} + (\gamma - \delta)\sigma^2. \end{aligned}$$

The latter is greater than the former when

$$\frac{\beta_1^2}{4\gamma} + (\gamma - \delta)\sigma^2 \geq \frac{\beta_1^2}{4\delta} \iff (\gamma - \delta)\sigma^2 \geq \frac{\beta_1^2}{4} \left(\frac{1}{\delta} - \frac{1}{\gamma} \right) \iff \sigma^2 \geq \frac{\beta_1^2}{\gamma\delta}.$$

Now, suppose $\delta \in \theta_i^\circ, \theta_j^\circ$. Then, the receiver entertains just the sender's model, which she believes is the sender's (only) model. Thus, we consider situations where the boundaries of the sets intersect.

Next, suppose $\underline{\delta}_i \in \theta_j^\circ$. Conditional on seeing $\underline{\delta}_i$, the receiver entertains models with $\gamma \leq \underline{\delta}_i$. By Theorem 1, the sender would prefer to locally sample in these cases. However, in his initial choice of σ , the sender is forced to consider both the probability of $\bar{\delta}_i$ and $\underline{\delta}_i$. If $\sigma_i = \bar{\sigma}$, the sender is being forced to distantly sample, which delivers a gain most of the time but a loss in the case where $\underline{\delta}_i$ materializes. However, if $\underline{\delta}_i \in \theta_j^\circ$, then with some probability, the receiver believes the $\underline{\delta}_i$ model was the true model and came from the type j sender. This means the receiver places a higher weight on the sender's model itself, and less on models with lower γ values (which, under $\sigma_i = \bar{\sigma}$, deliver lower expected utility).

A similar argument shows that the sender *loses out* when $\delta = \underline{\delta}_i$ and $\sigma_i = \underline{\sigma}$. In this case, the sender is (optimally) gaining from his own local sampling, meaning the remaining models the receiver entertains deliver higher expected utility. However, the receiver thinks with positive probability that the sender's model is in fact the *true* model, causing her to relatively downweight these alternative models and focus more weight on the sender's model, which delivers lower expected utility. Yet another similar argument shows that the sender gains (loses) when $\bar{\delta}_i \in \theta_j^\circ$ and $\sigma_i = \underline{\sigma}$ ($\sigma_i = \bar{\sigma}$).

Next, suppose $\delta \in \theta_i^\circ$ and that $\delta = \bar{\delta}_j$. If the type j sender is distantly sampling ($\sigma_j = \bar{\sigma}$), the receiver entertains not only type i 's model but also models with higher γ (since she believes the model could have also come from θ_j). The set of models R entertains is thus more optimistic than if she were to just entertain the type i sender's model, meaning the type i sender benefits from the type j 's distant sampling. Local sampling, on the other hand, would hurt him. A similar argument takes care of the case where $\delta = \underline{\delta}_j$, where the type θ_i sender is hurt by distant sampling and benefits from local sampling.

□

Theorem 4. Suppose P_δ^D is continuous. Then, the optimal type space is the collection of singletons on

$[0, \bar{\gamma}] = \Gamma: \Theta = \{\{\delta\} : \delta \in \Gamma\}$. Moreover, there exists a function $\pi(\delta)$ such that T is given by a continuous distribution t with

$$t(\delta) = p_\delta^D(\delta)\pi(\delta). \quad (T^*)$$

In particular, there exist thresholds $0 \leq \underline{\delta}_0 < \mathbb{E}_\gamma^R[\gamma] < \bar{\delta}_0 \leq \bar{\gamma}$ such that $\pi(\delta)$ is strictly decreasing below $\underline{\delta}_0$, strictly increasing above $\bar{\delta}_0$, and equal to 0 on $[\underline{\delta}_0, \bar{\delta}_0]$.

Proof. Consider a sender model $f_S = \beta_0 + \beta_1 x - \delta x^2$. First, we show that Θ contains no non-degenerate intervals. Suppose by contradiction that there existed an interval of type $\theta = [\underline{\delta}_\theta, \bar{\delta}_\theta]$ with $\theta \subseteq \Gamma^\circ$. Suppose $\theta^\circ \neq \emptyset$ and consider $\delta \in \theta^\circ$. Conditional on seeing this model, the receiver believes the model $f_S(x) = \beta_0 + \beta_1 x - \delta x^2$ is the true model.

Now, for each $\delta \in \theta^\circ$, as before, D can create a type $\{\delta\}$. This type strategically samples by setting σ such that $\max_x f_S(x) \leq \max_x f_S(x) + (\mathbb{E}[\gamma|\delta])(\sigma^2 - (x - x_\theta)^2)$. Thus, conditional on seeing a model of the form $\beta_0 + \beta_1 x - \delta x^2$, R believes that $f_S(x)$ is either the true model or the model from the $\{\delta\}$ type. Let $q(\delta|\beta_0, \beta_1, T)$ be the conditional probability of the type $\{\delta\}$ sender relative to the original type θ sender. Let $\bar{\gamma}_{\{\delta\}}$ be the receiver's expectation of γ if $\delta \geq \gamma$ given f_S and $\bar{\gamma}_{\{\delta\}}$ likewise if $\delta < \gamma$. Then, this generates a difference on the order of

$$\begin{aligned} & \int_{\underline{\delta}_\theta}^{\bar{\delta}_\theta} \max_x f_S(x) + q(\delta|\beta_0, \beta_1, T)(\bar{\gamma}_{\{\delta\}} - \delta)(\bar{\sigma}^2 - (x - x_\theta)^2) - \max_x f_S(x) dP_\delta^D(\delta \leq \mathbb{E}_\gamma^R[\gamma] | \delta \in \theta^\circ) \\ & + \int_{\underline{\delta}_\theta}^{\bar{\delta}_\theta} \max_x f_S(x) + q(\delta|\beta_0, \beta_1, T)(\underline{\gamma}_{\{\delta\}} - \delta)(\underline{\sigma}^2 - (x - x_\theta)^2) - \max_x f_S(x) dP_\delta^D(\delta > \mathbb{E}_\gamma^R[\gamma] | \delta \in \theta^\circ) \end{aligned}$$

The first line considers any arbitrary senders who position as generalists. The second line considers any arbitrary senders who position as specialists. In both cases, the gain is positive and is increasing in $q(\delta|\beta_0, \beta_1, T)$; i.e., the gain is higher the more likely R is to think that $f_S(x)$ came from type $\{\delta\}$.

We are left to consider the boundary types of θ : $\bar{\delta}_\theta$ and $\underline{\delta}_\theta$. Since θ either samples as a generalist or specialist, by Proposition 8 one of $\bar{\delta}_\theta$ or $\underline{\delta}_\theta$ gains and one loses from the existence of an arbitrary type. However, because P_δ^D is continuous, these singletons have measure 0 with respect to the prior. Thus, replacing each with an arbitrary type has no effect on expected utility.

Hence, every type space Θ with non-degenerate intervals can be improved upon by replacing all the senders in that interval with singletons. The maximal set of singletons covers all of Γ , meaning $\Theta = \{\{\delta\} : \delta \in \Gamma\}$ maximizes the objective.

We now determine the distribution of types T . Fix a model $\beta_0 + \beta_1 x - \delta x^2$. First, for a distribution $T \in \mathcal{P}([0, \bar{\gamma}])$ (the set of measures over $[0, \bar{\gamma}]$), we write as $q(T, \delta)$ the posterior probability that this model came from a type $\{\delta\}$ sender.

$$q(T, \delta) = \begin{cases} \frac{(1-q_0)t(\delta)}{(1-q_0)t(\delta) + q_0 p_\delta^D(\delta)} & T \text{ admits density } t(\delta) \\ 1 & T(\{\delta\}) \text{ has a mass point} \end{cases}$$

With some abuse of notation we write as $\mathbb{E}_R^\gamma[\gamma]$ R 's expectation given the sender was a simple type who sampled at $x_{\{\delta\}}$. meaning that

$$\mathbb{E}_\gamma^R[\gamma|f_S] = q(T, \delta)\mathbb{E}_\gamma^R[\gamma] + (1 - q(T, \delta))\delta.$$

Given a model $f_S(x) = \beta_0 + \beta_1 x - \delta x^2$, the receiver's maximal expected value is

$$f_S(x^*) + q(T, \delta)(\mathbb{E}_\gamma^R[\gamma] - \delta)(\sigma_\delta^2 - (x^* - x_\delta)^2)$$

where x^* is a function of β_1 , δ , and $\mathbb{E}_\gamma^R[\gamma|f_S]$, and σ_δ and x_δ are determined as in Theorem 1.

Our objective can thus be rewritten as:

$$\max_{T \in \mathcal{P}([0, \bar{\gamma})]} \int_{\mathbb{R}} \int_{\mathbb{R}} \int_0^{\bar{\gamma}} f_S(x^*) + q(T, \delta)(\mathbb{E}_\gamma^R[\gamma] - \delta)(\sigma_\delta^2 - (x^* - x_\delta)^2) dP_\delta^D(\delta) dP_1^D(\beta_1) dP_0^D(\beta_0),$$

Note that because P_δ^D is continuous, there are no gains to placing a positive mass on any singleton δ value. Hence, our problem boils down to

$$\begin{aligned} \max_{t(\delta)} \int_{\mathbb{R}} \int_{\mathbb{R}} \int_0^{\bar{\gamma}} f_S(x^*) + q(T, \delta)(\mathbb{E}_\gamma^R[\gamma] - \delta)(\sigma_\delta^2 - (x^* - x_\delta)^2) dP_\delta^D(\delta) dP_1^D(\beta_1) dP_0^D(\beta_0) \\ \text{s.t. } t(\delta) \geq 0 \text{ and } \int_0^{\bar{\gamma}} t(\delta) d\delta = 1. \end{aligned}$$

Using a calculus of variations approach, we can write a Lagrangian as:

$$\mathcal{L}(t, \lambda) = \int_{\mathbb{R}} \int_{\mathbb{R}} \int_0^{\bar{\gamma}} f_S(x^*) + q(T, \delta)(\mathbb{E}_\gamma^R[\gamma] - \delta)(\sigma_\delta^2 - (x^* - x_\delta)^2) p_\delta^D(\delta) + \lambda \left(\frac{1}{\bar{\gamma}} - t(\delta) \right) d\delta dP_1^D(\beta_1) dP_0^D(\beta_0).$$

Using the envelope theorem, we obtain the following first-order condition.

$$\begin{aligned} \mathbb{E}_{0,1}^D \left[\frac{(1 - q_0)((1 - q_0)t(\delta) + q_0 p_\delta^D(\delta)) - (1 - q_0)^2 t(\delta)}{((1 - q_0)t(\delta) + q_0 p_\delta^D(\delta))^2} (\mathbb{E}_\gamma^R[\gamma] - \delta)(\sigma_\delta^2 - (x^* - x_\delta)^2) p_\delta^D(\delta) \right] - \lambda \\ = \frac{(1 - q_0) q_0 p_\delta^D(\delta)}{((1 - q_0)t(\delta) + q_0 p_\delta^D(\delta))^2} p_\delta^D(\delta) (\mathbb{E}_\gamma^R[\gamma] - \delta)(\sigma_\delta^2 - \mathbb{E}_1^D[(x^* - x_\delta)^2]) - \lambda, \end{aligned}$$

where we note that x^* depends on β_1 and δ . Now, define

$$P = q_0 p_\delta^D(\delta) \quad R = (1 - q_0)t(\delta).$$

Recall that $x^* = x_\delta + \frac{\beta_1 - 2\delta x_\delta}{2\mathbb{E}_\gamma^R[\gamma|\delta]}$. We then have

$$\begin{aligned} \frac{(1-q_0)P^2}{q_0(P+R)^2}(\mathbb{E}_\gamma^R[\gamma] - \delta)(\sigma_\delta^2 - \mathbb{E}_1^D[(x^* - x_\delta)^2]) &= \frac{(1-q_0)P^2}{q_0(P+R)^2}(\mathbb{E}_\gamma^R[\gamma] - \delta)(\sigma_\delta^2 - \mathbb{E}_1^D[(\frac{\beta_1 - 2\delta x_\delta}{2\mathbb{E}_\gamma^R[\gamma|\delta]})^2]) \\ &= \frac{(1-q_0)P^2}{q_0(P+R)^2}(\mathbb{E}_\gamma^R[\gamma] - \delta)(\sigma_\delta^2 - \frac{\mathbb{E}_1^D[(\beta_1 - 2\delta x_\delta)^2]}{\frac{4(P\delta + R\mathbb{E}_\gamma^R[\gamma])^2}{(P+R)^2}}) \\ &= \frac{(1-q_0)}{q_0}P^2(\mathbb{E}_\gamma^R[\gamma] - \delta)(\frac{\sigma_\delta^2}{(P+R)^2} - \frac{\mathbb{E}_1^D[(\beta_1 - 2\delta x_\delta)^2]}{4(P\delta + R\mathbb{E}_\gamma^R[\gamma])^2}). \end{aligned}$$

Next, define $\pi(\delta) := \frac{t(\delta)}{p_\delta^D(\delta)}$ and $c = \frac{1-q_0}{q_0}$. In particular, $\frac{R}{P} = c\pi(\delta)$. This definition gives:

$$\begin{aligned} \frac{(1-q_0)}{q_0}P^2(\mathbb{E}_\gamma^R[\gamma] - \delta)(\frac{\sigma_\delta^2}{(P+R)^2} - \frac{\mathbb{E}_1^D[(\beta_1 - 2\delta x_\delta)^2]}{4(P\delta + R\mathbb{E}_\gamma^R[\gamma])^2}) \\ = c(\mathbb{E}_\gamma^R[\gamma] - \delta)(\frac{\sigma_\delta^2}{(1+c\pi(\delta))^2} - \frac{\mathbb{E}_1^D[(\beta_1 - 2\delta x_\delta)^2]}{4(\delta + c\pi(\delta)\mathbb{E}_\gamma^R[\gamma])^2}). \end{aligned}$$

Finally, we verify the second order condition; because $\pi(\delta) = \frac{t(\delta)}{p_\delta^D(\delta)}$, differentiating with respect to t yields:

$$c^2 p_\delta^D(\delta)(\mathbb{E}_\gamma^R[\gamma] - \delta)(-\frac{\sigma_\delta^2}{(1+c\pi(\delta))^3} + \mathbb{E}_\gamma^R[\gamma] \frac{\mathbb{E}_1^D[(\beta_1 - 2\delta x_\delta)^2]}{2(\delta + c\pi(\delta)\mathbb{E}_\gamma^R[\gamma])^3}).$$

For $\delta < \mathbb{E}_\gamma^R[\gamma]$, we note that $\sigma_\delta \rightarrow \bar{\sigma}$, meaning $-\frac{\sigma_\delta^2}{(1+c\pi(\delta))^3} + \mathbb{E}_\gamma^R[\gamma] \frac{\mathbb{E}_1^D[(\beta_1 - 2\delta x_\delta)^2]}{2(\delta + c\pi(\delta)\mathbb{E}_\gamma^R[\gamma])^3} < 0$. For $\delta > \mathbb{E}_\gamma^R[\gamma]$, $\sigma_\delta \rightarrow \underline{\sigma}$ small, meaning $-\frac{\sigma_\delta^2}{(1+c\pi(\delta))^3} + \mathbb{E}_\gamma^R[\gamma] \frac{\mathbb{E}_1^D[(\beta_1 - 2\delta x_\delta)^2]}{2(\delta + c\pi(\delta)\mathbb{E}_\gamma^R[\gamma])^3} > 0$, and thus that the whole expression is negative. Hence, the original objective is concave in t , meaning the first order condition identifying a solution is given by

$$c(\mathbb{E}_\gamma^R[\gamma] - \delta)(\frac{\sigma_\delta^2}{(1+c\pi(\delta))^2} - \frac{\mathbb{E}_1^D[(\beta_1 - 2\delta x_\delta)^2]}{4(\delta + c\pi(\delta)\mathbb{E}_\gamma^R[\gamma])^2}) = \lambda.$$

Note that the left hand side is strictly positive except at $\delta = \mathbb{E}_\gamma^R[\gamma]$, when it is equal to 0. We define:

$$\xi(\delta, \pi) := c(\mathbb{E}_\gamma^R[\gamma] - \delta)(\frac{\sigma_\delta^2}{(1+c\pi)^2} - \frac{\mathbb{E}_1^D[(\beta_1 - 2\delta x_\delta)^2]}{4(\delta + c\pi\mathbb{E}_\gamma^R[\gamma])^2}).$$

By the Implicit Function Theorem, we have $\pi'(\delta) = -\frac{\xi_\pi}{\xi_\delta}$. But note that $\xi_\pi < 0$ (since the objective was concave in t , and $t = p_\delta^D(\delta) \cdot \pi$), meaning that the sign of π' is just the sign of ξ_δ .

Now, consider $\delta < \mathbb{E}_\gamma^R[\gamma]$. These senders sample so that $\sigma_\delta = \bar{\sigma}$ and $x_\delta = \max\{0, \min\{\frac{\mathbb{E}_1^D[\beta_1]}{2\delta}, \bar{x}\}\}$.

Consider first the interior case, which gives

$$\begin{aligned}\mathbb{E}_1^D[(\beta_1 - 2\delta x_\delta)^2] &= \mathbb{E}_1^D[(\beta_1 - 2\delta \frac{\mathbb{E}_1^D[\beta_1]}{2\delta})^2] = \text{Var}_1^D[\beta_1] \\ \implies \xi(\delta, \pi) &= c(\mathbb{E}_\gamma^R[\gamma] - \delta) \left(\frac{\sigma_\delta^2}{(1 + c\pi)^2} - \frac{\text{Var}_1^D[\beta_1]}{4(\delta + c\pi \mathbb{E}_\gamma^R[\gamma])^2} \right).\end{aligned}$$

The derivative of this expression with respect to δ is < 0 if and only if

$$\begin{aligned}-c \left(\frac{\sigma_\delta^2}{(1 + c\pi)^2} - \frac{\text{Var}_1^D[\beta_1]}{4(\delta + c\pi \mathbb{E}_\gamma^R[\gamma])^2} \right) + c(\mathbb{E}_\gamma^R[\gamma] - \delta) \frac{\text{Var}_1^D[\beta_1]}{2(\delta + c\pi \mathbb{E}_\gamma^R[\gamma])^3} &< 0 \\ \frac{\sigma_\delta^2}{(1 + c\pi)^2} &\geq \frac{\text{Var}_1^D[\beta_1]}{4(\delta + c\pi \mathbb{E}_\gamma^R[\gamma])^2} \left(1 + \frac{2(\mathbb{E}_\gamma^R[\gamma] - \delta)}{\delta + c\pi \mathbb{E}_\gamma^R[\gamma]} \right)\end{aligned}$$

which always holds for $\sigma_\delta \rightarrow \bar{\sigma}$ sufficiently large. We are left to consider the case of a corner solution (0 or $\bar{x}$). The derivative ξ with respect to δ in this case is < 0 if and only if

$$-\left(\frac{\sigma_\delta^2}{(1 + c\pi)^2} - \frac{\mathbb{E}_1^D[(\beta_1 - 2\delta x_\delta)^2]}{4(\delta + c\pi \mathbb{E}_\gamma^R[\gamma])^2} \right) + (\mathbb{E}_\gamma^R[\gamma] - \delta) \left(\frac{\mathbb{E}_1^D[(\beta_1 - 2\delta x_\delta)^2]}{2(\delta + c\pi \mathbb{E}_\gamma^R[\gamma])^3} + \frac{x_\delta \mathbb{E}_1^D[(\beta_1 - 2\delta x_\delta)]}{(\delta + c\pi \mathbb{E}_\gamma^R[\gamma])^2} \right) < 0,$$

which again holds for σ_δ^2 large. Thus, $\pi(\delta)$ is decreasing in δ for $\delta < \mathbb{E}_\gamma^R[\gamma]$.

Now, we consider $\delta > \mathbb{E}_\gamma^R[\gamma]$. In this case, we have $\sigma_\delta \rightarrow 0$. Moreover, we have that $x_\delta = \arg \max_{x \in [0, \bar{x}]} \mathbb{E}_1^D[(\beta_1 - 2\delta x)^2]$ (i.e. specialist senders sample wherever they expect the curve to be steepest). In this case, we have

$$\xi(\delta, \pi) \approx c(\delta - \mathbb{E}_\gamma^R[\gamma]) \frac{\mathbb{E}_1^D[(\beta_1 - 2\delta x_\delta)^2]}{4(\delta + c\pi \mathbb{E}_\gamma^R[\gamma])^2}.$$

$\delta - \mathbb{E}_\gamma^R[\gamma]$ is increasing in δ (and positive for $\delta > \mathbb{E}_\gamma^R[\gamma]$). Using the fact that x_δ maximizes $\mathbb{E}_1^D[(\beta_1 - 2\delta x)^2]$, the second term is also increasing in δ for x_δ extreme.

Thus, whenever $\pi(\delta) > 0$, it is strictly decreasing for $\delta < \mathbb{E}_\gamma^R[\gamma]$ and strictly increasing for $\delta > \mathbb{E}_\gamma^R[\gamma]$. To see that $\pi(\delta) = 0$ on an interior interval we simply note that for every $\lambda > 0$, there exists an interval of δ values containing $\mathbb{E}_\gamma^R[\gamma]$ such that

$$c(\mathbb{E}_\gamma^R[\gamma] - \delta) \left(\frac{\sigma_\delta^2}{(1 + c\pi(\delta))^2} - \frac{\mathbb{E}_1^D[(\beta_1 - 2\delta x_\delta)^2]}{4(\delta + c\pi(\delta) \mathbb{E}_\gamma^R[\gamma])^2} \right) < \lambda$$

at $\pi = 0$ (and, by concavity, for all $\pi \geq 0$).

□

Proposition 9. *The polarization of the type space is increasing in q_0 . That is, $\underline{\delta}_0$ is decreasing in q_0 and $\bar{\delta}_0$ is increasing in q_0 . As $q_0 \rightarrow 1$, $t(\delta)$ converges to a distribution that either places all mass only on $\delta = 0$, $\delta = \bar{\gamma}$, or both. As $q_0 \rightarrow 0$, $t(\delta)$ converges to a distribution that is strictly decreasing on $[0, \mathbb{E}_\gamma^R[\gamma])$, equal to 0 at $\mathbb{E}_\gamma^R[\gamma]$, and strictly increasing on $(\mathbb{E}_\gamma^R[\gamma], \bar{\gamma}]$.*

Proof. We recall our initial first order condition implicitly defining $\pi(\delta)$

$$c(\mathbb{E}_\gamma^R[\gamma] - \delta) \left(\frac{\sigma_\delta^2}{(1 + c\pi(\delta))^2} - \frac{\mathbb{E}_1^D[(\beta_1 - 2\delta x_\delta)^2]}{4(\delta + c\pi(\delta)\mathbb{E}_\gamma^R[\gamma])^2} \right) = \lambda,$$

where the left hand side is decreasing in π , decreasing in δ for $\delta < \mathbb{E}_\gamma^R[\gamma]$, and increasing in δ for $\delta > \mathbb{E}_\gamma^R[\gamma]$. We also recall that $c := \frac{1-q_0}{q_0}$, and that as $c \rightarrow 0$, $q_0 \rightarrow 1$, and that as $c \rightarrow \infty$, $q_0 \rightarrow 0$.

Recall also that $\bar{\delta}_0$ and $\underline{\delta}_0$ are the points at which $\pi(\delta) = 0$ and are, in turn, functions of c . By the constraint that $t(\delta)$ is a density and by writing $\pi(\delta, c)$ as a function of c , we have

$$\begin{aligned} \int_0^{\bar{\gamma}} t(\delta) d\delta &= \int_0^{\underline{\delta}_0(c)} \pi(\delta, c) p_\delta^D(\delta) d\delta + \int_{\bar{\delta}_0(c)}^{\bar{\gamma}} \pi(\delta, c) p_\delta^D(\delta) d\delta = 1 \\ \pi(\bar{\delta}_0(c), c) p_\delta^D(\bar{\delta}_0(c)) \bar{\delta}_0'(c) - \pi(\underline{\delta}_0(c), c) p_\delta^D(\underline{\delta}_0(c)) \underline{\delta}_0'(c) &+ \int_0^{\underline{\delta}_0(c)} \pi_c(\delta, c) p_\delta^D(\delta) + \int_{\bar{\delta}_0(c)}^{\bar{\gamma}} \pi_c(\delta, c) p_\delta^D(\delta) \\ &= \int_0^{\bar{\gamma}} \pi_c(\delta, c) p_\delta^D(\delta) d\delta = 0, \end{aligned}$$

where we use the fact that $\pi(\bar{\delta}_0(c), c) = \pi(\underline{\delta}_0(c), c) = 0$.

Next, note that, fixing δ , the first order condition can be rearranged as

$$\Xi(c\pi, \delta) := (\mathbb{E}_\gamma^R[\gamma] - \delta) \left(\frac{\sigma_\delta^2}{(1 + c\pi)^2} - \frac{\mathbb{E}_1^D[(\beta_1 - 2\delta x_\delta)^2]}{4(\delta + c\pi\mathbb{E}_\gamma^R[\gamma])^2} \right) = \frac{\lambda(c)}{c}$$

The FOC then gives

$$\Xi_{(c\pi)}(c\pi_c + \pi) = \frac{\lambda'(c)}{c} - \frac{\lambda(c)}{c^2} \implies c\pi_c = \frac{\frac{\lambda'(c)}{c} - \frac{\lambda(c)}{c^2}}{\Xi_{(c\pi)}} - \pi$$

But then, using the boundary condition, we have

$$\begin{aligned} c \int_0^{\bar{\gamma}} \pi_c(\delta, c) p_\delta^D(\delta) d\delta &= \int_0^{\bar{\gamma}} \frac{\frac{\lambda'(c)}{c} - \frac{\lambda(c)}{c^2}}{\Xi_{(c\pi)}} p_\delta^D(\delta) - \pi(\delta, c) p_\delta^D(\delta) d\delta = 0 \\ &\implies \left(\lambda'(c) - \frac{\lambda(c)}{c} \right) \int_0^{\bar{\gamma}} \frac{1}{\Xi_{(c\pi)}(c\pi, \delta)} p_\delta^D(\delta) d\delta = c \\ &\implies \frac{\lambda'(c)}{c} - \frac{\lambda(c)}{c^2} = \frac{1}{\int_0^{\bar{\gamma}} \frac{1}{\Xi_{(c\pi)}(c\pi, \delta)} p_\delta^D(\delta) d\delta} < 0 \end{aligned}$$

since $\Xi_{(c\pi)} < 0$ (i.e. Ξ is decreasing in $c\pi$, from the proof of the previous Theorem).

Now, consider $\Xi(0, \underline{\delta}_0(c))$, which implicitly defines $\underline{\delta}_0(c)$ as

$$\Xi(0, \underline{\delta}_0(c)) := (\mathbb{E}_\gamma^R[\gamma] - \underline{\delta}_0(c)) \left(\sigma_\delta^2 - \frac{\mathbb{E}_1^D[(\beta_1 - 2\underline{\delta}_0(c)x_\delta)^2]}{4(\underline{\delta}_0(c))^2} \right) = \frac{\lambda(c)}{c}.$$

In this case, we have

$$\Xi_\delta(0, \underline{\delta}_0(c)) \underline{\delta}'_0(c) = \frac{\lambda'(c)}{c} - \frac{\lambda(c)}{c^2} < 0.$$

But recall from the proof of the previous result that $\Xi_\delta < 0$ for $\delta < \mathbb{E}_\gamma^R[\gamma]$. This means that $\underline{\delta}_0(c)$ is increasing in c , i.e. is increasing in $\frac{1-q_0}{q_0}$. Hence, as q_0 increases, $\underline{\delta}_0$ decreases. An analogous proof shows that as q_0 increases, $\bar{\delta}_0$ increases. That is, the “polarization” induced by π is increasing in q_0 .

Finally, we investigate the limiting cases as $q_0 \rightarrow 0$ ($c \rightarrow \infty$) and $q_0 \rightarrow 1$ ($c \rightarrow 0$). We begin with $c \rightarrow \infty$, and claim that $\underline{\delta}_0 \rightarrow \mathbb{E}_\gamma^R[\gamma]$. First, by our work above $\frac{\lambda(c)}{c}$ is decreasing in c , meaning it converges to a limit. We claim $\lim_{c \rightarrow \infty} \frac{\lambda(c)}{c} = 0$. Suppose by contradiction that this did not hold, and instead it converged to some $k > 0$. This would mean that there exists $\bar{c\pi} > 0$ such that $c\pi(\delta) \leq \bar{c\pi}$ for all c and $\delta \in [0, \bar{\gamma}] - \lim_{c \rightarrow \infty} [\underline{\delta}_0(c), \bar{\delta}_0(c)]$. But then, for all c we would have

$$\int_0^{\bar{\gamma}} \pi(\delta, c) p_\delta^D(\delta) d\delta \leq \int_0^{\bar{\gamma}} \bar{c\pi} p_\delta^D(\delta) d\delta = \frac{\bar{c\pi}}{c},$$

which approaches 0 as $c \rightarrow \infty$, contradicting that $\int_0^{\bar{\gamma}} \pi(\delta, c) p_\delta^D(\delta) d\delta = 1$. Hence, we have that $\lim_{c \rightarrow \infty} \frac{\lambda(c)}{c} = 0$, meaning $\lim_{c \rightarrow \infty} \underline{\delta}_0(c) = \lim_{c \rightarrow \infty} \bar{\delta}_0(c) = \mathbb{E}_\gamma^R[\gamma]$. Moreover, we note that even for c arbitrarily large, $\pi(\mathbb{E}_\gamma^R[\gamma]) = 0$, with $\pi'(\delta) < 0$ below $\mathbb{E}_\gamma^R[\gamma]$ and $\pi'(\delta) > 0$ above $\mathbb{E}_\gamma^R[\gamma]$ since $\Xi(c\pi, \delta)$ is decreasing in δ and increasing in δ on each range, respectively.

Finally, we investigate the case where $c \rightarrow 0$. We claim that as $c \rightarrow 0$, $\underline{\delta}_0 \rightarrow 0$ and $\bar{\delta}_0 \rightarrow \bar{\gamma}$ (and hence t converges to a discrete distribution that either concentrates mass on 0 or $\bar{\gamma}$ or both). In particular, note that for all c , we have

$$c = \int_0^{\bar{\gamma}} c\pi(\delta, c) p_\delta^D(\delta) d\delta,$$

which means that as $c \rightarrow 0$, we necessarily must have $c\pi(\delta, c) \rightarrow 0$ as well. This can happen only if

$$(\mathbb{E}_\gamma^R[\gamma] - \delta) \left(\sigma_\delta^2 - \frac{\mathbb{E}_1^D[(\beta_1 - 2\delta x_\delta)^2]}{4(\delta)^2} \right) \leq \lim_{c \rightarrow 0} \frac{\lambda(c)}{c},$$

i.e. either the multiplier constraint must hold with equality or fail to hold (generating a corner solution at $\pi(\delta, c) = 0$). Since this must hold pointwise for each δ for c sufficiently small, we thus have that $\underline{\delta}_0 \rightarrow 0$ and $\bar{\delta}_0 \rightarrow \bar{\gamma}$, giving the result. $\square$